46.40 M FLAT TOP PONTOON

INTACT STABILITY BOOKLET

0	17-May-18	Initial issue for approval	AJAY	vissu	SC
Rev	Date	Description	Prepared	Checked	Reviewed

Client:

ARCADIA SHIPPING AND TRADING CO.

Title:

INTACT STABILITY BOOKLET

46.40 M FLAT TOP PONTOON

Date: 17-May-19	Sheet: Page **1** of **45**	Project: 18-1018
Document: SBM-18-1018-305		Rev: 0

SB Marine
Marine and Offshore Structure Consultant
Tel: +91 22 25970189
Web: www.sbmarineconsultant.com

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1 INTRODUCTION

This booklet consists of intact stability analysis of the vessel "46.4 M FLAT TOP PONTOON" used for carrying deck cargo or operation of crane with a boom length of 40.0 m and a hook load of 45 MT at 16 m operating radius located at center of the pontoon. The pontoon is loaded with other deck equipment as shown in the loading conditions. The pontoon is complying the necessary stability requirements as per stability criteria given in section 3.

1.1 VESSEL PARTICULARS

Ship's Name	:	"46.4 M FLAT TOP PONTOON"
Client's Name	:	ARCADIA SHIPPING AND TRADING CO
Length Overall	:	46.40 Meter
Breadth Moulded	:	18.00 Meter
Depth Moulded	:	3.0 Meter (above Base Line)
Designed Draft	:	2.20 meter (above baseline)
Maximum Draft Moulded	:	1.105 Meter (above Base Line)

2 NOTES ON THE USE OF STABILITY BOOKLET

2.1 CONVERSION TABLES

The Government of India strongly recommends use of S.I (Systems International Units). The following table shows conversion factors for converting British Units to Metric Units and vice versa.

Multiply by	To Convert from	To Obtain	
0.03937	Millimeters	Inches	25.4
0.3937	Centimeters	Inches	2.54
3.2808	Meters	Feet	0.3048
2.2046	Kilograms	Pound	0.45359
0.0009842	Kilograms	Tons (2240 lbs.)	1016.047
0.9842	Metric Tonnes	Tones (2240 lbs.)	1.016
2.4998	Metric Tonne per centimeter of immersion	Tones per inch of immersion	0.04
8.2014	Moment to change trim by one centimeter (tonne-metre)	Moment to change trim by one inch (Ton-foot)	0.122
0.0174532	Degree	Radians	57.2957795
	TO OBTAIN	**TO CONVERT FROM**	**MULTIPLY BY**

Relation between Weight and Volume:

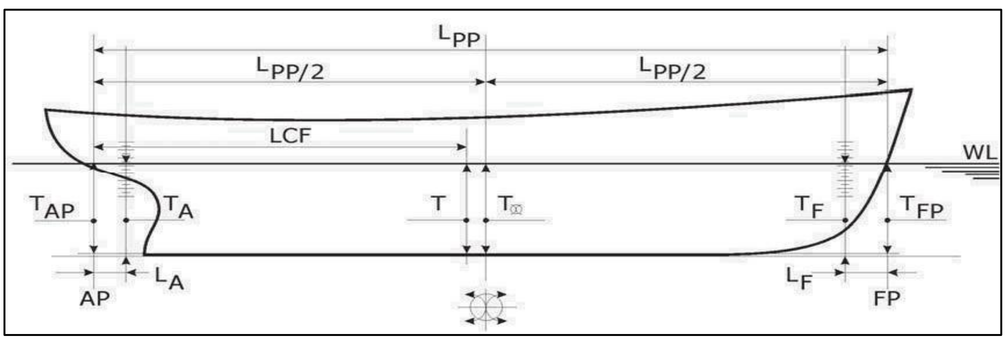
10^3 mm	=	1 cm^3
1 cm^3 of fresh water	=	1 Gram
1000 cm^3 of fresh water	=	1 Kilogram (1000 Grams)
1 m^3 of fresh water	=	1 Metric tonne (1000 Kilograms)
1 m^3 of salt water (Specific Gravity = 1.025 t/m^3)	=	1.025 Metric Tonnes
1 Metric Tonne of salt water	=	0.975 m^3
1 m^3	=	35.316 ft^3
1 ft^3	=	0.0283 m^3

2.2 SYMBOLS AND SIGN CONVENTION

SYMBOL	DESCRIPTION	UNIT
Δ	Displacement	t
$\bar{\nabla}$	Volume of Displacement	m^3
AP	Aft perpendicular	m
A_{WP}	Area of Water Plane	m^2
B	Breadth Moulded	m
D	Depth Moulded (Up to Main Deck at Side)	m
Dwt.	Deadweight	tonnes
FP	Forward perpendicular	m
L_A	Distance from AP to aft draft mark	m
	(Positive value if draft mark is positioned forward of AP)	
LBP	Length between perpendiculars	m
LCB	Longitudinal Centre of Buoyancy (from transom)	m
LCF	Longitudinal Centre of Flotation (from transom)	m
LCG	Longitudinal center of gravity (from transom)	m
L_F	Distance from FP to forward draft mark	m
	(Positive value if draft mark is positioned forward of FP)	
TCG	Transverse center of gravity from center line	m
GG_O	Free surface correction	m
GM_T	Transverse metacentric height	m
G_OM_T	Corrected transverse metacentric height	m
GM_L	Longitudinal metacentric height	m
GZ	Righting arm	m
GT	Gross Tonnage	-
G_oZ	Corrected righting arm	m
G_oZ_{Max}	Corrected maximum righting arm	m
KB	Vertical Centre of Buoyancy (from Base Line)	m
KG/VCG	Vertical center of gravity from base line	m
KG_O	Corrected vertical center of gravity from base line	m

KM_T	Transverse Metacentric Height (from Base Line)	m
KN	Distance from base line to metacentric axis	m
MCT_{lcm}	Moment to change trim by one centimeter	tones x m/cm
NT	Net Tonnage	-
TPC	Tones per centimeter immersion	tonnes/cm
t	Total Trim (T_{FP} - T_{AP})	m
T	Draft at LCF	m
T_{AP}	Draft Aft at AP	m
T_A	Draft aft draft marks from underside of keel	m
T_{FP}	Draft Forward at FP	m
T_F	Draft forward draft marks from underside of keel	m
T_M	Draft Mean at Amidships	m
W_{llool}	Wind heeling arm	m
θ	Angle of inclination in degrees	degrees
ρ	Density of liquid	tonnes/m^3
θ_{Max}	Angle at which maximum G_0Z occurs	degrees

2.3 SIGN CONVENTION



LCB, LCG, LCF	Forward of Transom is positive Aft of Transom is Negative
Trim by Aft	Negative
Trim by Forward	Positive

Towards Port positive and Starboard side is Negative

2.4 GENERAL INSTRUCTIONS TO MASTER AGAINST CAPSIZING

The fact that the vessel's stability has been calculated for several conditions of loading in which the vessel normally operates, and that such calculations have proven that the stability is good and in excess of the IMO requirements, and the fact that the calculations are checked for approval does not mean that the vessel is now immune against capsizing. A stamped and approved copy of this booklet must be on board at all times. The following measures should be considered as preliminary guidance on matters influencing safety and stability.

1. Compliance with the stability criteria does not ensure immunity against capsizing, regardless of the circumstances, or absolve the master from his responsibilities. Masters should therefore exercise prudence and good seamanship having regard to the season of the year weather forecasts and the navigational zone and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Before a voyage commences, care should be taken to ensure that the cargo and sizeable pieces of equipment have been properly stowed or lashed so as to minimize the possibility of both longitudinal and shifting, while at sea, under the effect of acceleration caused by rolling and pitching.
3. The stability criteria recommended as per IS Code 2008 and set minimum values, but no maximum values are recommended. It is advisable to avoid excessive valves of metacentric height, since these might lead to acceleration forces which could be prejudicial to the vessel, its complement, its equipment and to safe carriage of the cargo.
4. All doorways and other openings through which can enter into the hull or deckhouses, forecastle, etc., should be suitably closed in adverse weather conditions and accordingly all appliances for this purpose should be maintained on board and in good condition.
5. Weather tight and watertight hatches, doors, etc., should be kept closed during navigation, except when necessarily opened for the working of the vessel and should always be ready for immediate closure and be clearly marked to indicate that these fittings are to be kept closed except for access. All portable deadlights should be maintained in good condition and securely closed inn bad weather.
6. Any closing devices provided for vent pipes to fuel tanks should be secured in bad weather.
7. Reliance on automatic steering may be dangerous as this prevents ready changes to course which may be needed in bad weather.
8. In all conditions of loading necessary care should be taken to maintain a seaworthy freeboard.
9. In severe weather, the speed of the vessel should be reduced if excessive rolling, propeller emergence, shipping of water on deck of heavy slamming occurs. Six heavy slamming or 25 propeller emergences during 100 pitching motions should be considered dangerous.

10. Special attention should be paid when a vessel is sailing in following or quartering seas because dangerous phenomena such as parametric resonance, broaching to reduction of satiability on the wave crest, and excessive rolling may occur singularly in sequence or simultaneously in a multiple combination, creating a threat of capsize. Particularly dangerous is the situation when the wave length is of the order of 1.0-1.5 vessel's length. A vessel's speed and / or course should be altered appropriately to avoid the above mentioned phenomena.
11. Water trapping in deck wells should be avoided. If freeing ports are not sufficient for draining of the well, either speed of the vessel should be reduced, or course should be changed, or both. Freeing ports provided with closing appliances should always be capable of functioning and are not to be locked.
12. Masters should be aware that steep or breaking waves may occur in certain areas, or in certain wind and current combinations.

2.5 NOTES ON STABILITY

Stability

The stability of a vessel in general is its ability to maintain an upright position or to re-establish this after disturbing by an external force or moment. For an undamaged vessel at small angle of heel up to 5^0, it is sufficient to evaluate the stability in transverse direction. This depends on position of two points relative to each other, center of gravity (G) and transverse metacenter (M), see figure. 2.4.1. The metacentric height GM, the distance between the points G and M, means the stability at small angles and is given by the following equation:

GMT = KMT - KG.

The center of gravity of the vessel (KG) depends on distribution of cargo and liquids in tanks in the vessel. Vessel KG for a particular loading condition is obtained by taking vertical moments (sum of weight of an item x center of gravity of that item) and by division of total moments with the total weights.

The position of the transverse metacenter varies with the draft. For a certain draft corresponding to a certain displacement, the transverse metacentric position (M) above the keel point K is fixed and is denoted as KMT. The location of the metacenter has neither to do with the nature nor the distribution of cargo & other weights onboard. The transverse metacenter above keel, only dependent on the lines of the vessel, shall be obtained from hydrostatics curves or tables.

Cross curves of stability

Position of transverse metacenter (M) is not fixed as the angle of heel exceeds beyond $5^0$and GM is no longer a stability criterion for measuring the stability of the vessel. Therefore righting lever or arm (GZ) shall be used for this purpose.

These curves show the variation of the righting arms (GZ) with displacement and are drawn for various angles of inclination, each curve being drawn for a particular angle of inclination. When these curves are drawn for a fixed value of the vertical center of gravity position, they are called “GZ Curves”. When the vertical center of gravity is assumed to be zero, these curves are called “KN Curves” and the same are used to derive the statical stability curve.

When the angle of inclination exceeds small angles, 4^0 or 5^0 the intersection of the vertical through the center of buoyancy in the inclined condition with the center line of vessel, point “M” can no longer be regarded as a fixed point relative to the vessel. The quantum “GM” no longer a suitable criterion for measuring the stability of the vessel and it is usual to use the value of the righting arm “GZ” for this purpose. The Cross Curves of stability provide a means of presenting the righting arm “GZ” for any probable value of displacement and for several angles of heel. The cross curves of stability depends on the geometry of the vessel and is independent of the loading of the vessel.

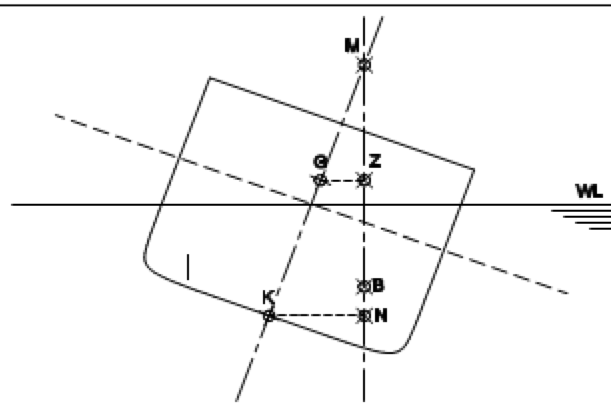
Curve of Statical Stability:

A curve of righting arm GZ to a base of angle of inclination for a fixed displacement is known as a curve of statical stability. This curve is provided for every intended loading condition in the booklet. To illustrate the righting levers at various angles of heel in a condition, the effective righting lever GZ is derived from:

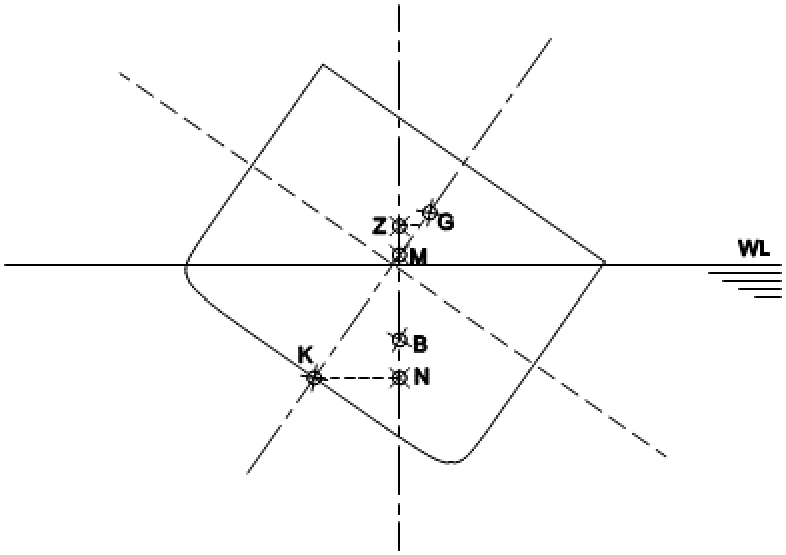
GZ	=	KN - KG x sin θ
KN	=	Horizontal distance from keel to the vertical line passing through center of Buoyancy.
KG	=	Centre of gravity from base line.
θ	=	Angle of inclination (heel).

Righting arm after correction of free surface effects is given by G_0Z.

G_0Z = KN - K_0G x sin θ



POSITIVE STABILITY



NEGATIVE STABILITY

Use of Tables:

The tables for hydrostatics, cross curves and maximum permissible KG should be entered with the ship's moulded draught amidships, T_M, measured from the upper side of the keel plate.

$$T_M = \frac{T_{AP} + T_{FP}}{2}$$

Draft and Trim Calculation:

For calculation of trim and draft based on a loading condition calculation, the following data are required for ascertaining trim and draft.

a) Longitudinal center of gravity of the ship calculated as follows.

$$LCG = \frac{SUMMATION\ OF\ LONGITUDINAL\ WEIGHT\ MOMENTS}{DISPLACEMENT}$$

b) Longitudinal center of buoyancy, LCB

c) Displacement, Δ

d) Moment to change trim, MCT

e) Longitudinal center of Floatation, LCF.

LCB, LCF and MCT are taken from Hydrostatic tables (trim = 0 meters) corresponding to the actual displacement.

Trim is calculated according to the following formula:

$$\text{Trim (t)} = \frac{LBP\ X\ (LCG - LCB)}{GM_L}$$

The Moulded Drafts at AP (T_{AP}) and FP (T_{FP}) are calculated to following formulas:

$$\text{Draft at LCF (m) } T_{LCF} = Tm - \left(\frac{\left(\frac{LBP}{2} - LCF\right) X\ TRIM}{LBP}\right)$$

$$T_{AP} = T_{LCF} - \frac{LCF\ X\ TRIM}{LBP}$$

$$T_{FP} = T_{AP} + TRIM$$

If the righting lever curve (GZ Curve) has to be established, the KM_T and KN values to be used for this purpose are calculated for the calculated trim value. This is done by interpolation for the KMT and KN values, respectively, between the two tabulated trim values nearest to calculated trim.

3 INTACT STABILITY CRITERIA

The vessel is analyzed to comply with the *intact stability criteria in accordance with Resolution MSC. 267(85) (IS Code 2008), Part B, Chapter 2.2.4.*

<u>Intact stability criteria as per regulation</u>

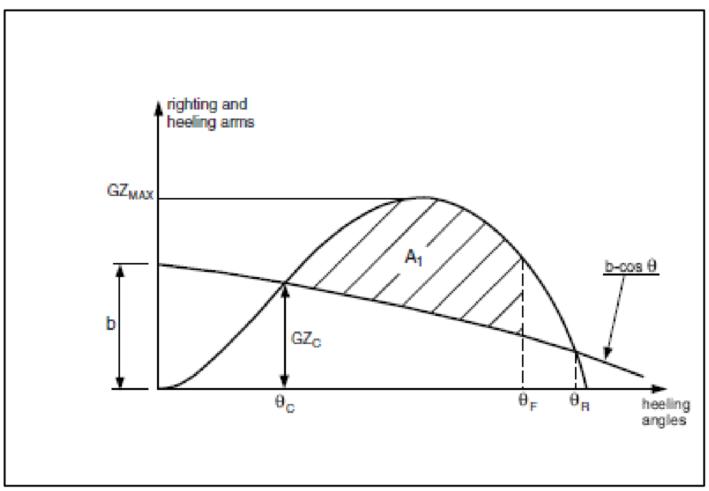
1. The area under the righting lever curve up to the angle of maximum righting lever should not be less than 0.08 meter-radians.
2. The static angle of heel due to a uniformly distributed wind load of 540 Pa (wind speed 30 m/s) should not exceed an angle corresponding to half the freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the Centroid of the Windage area to half the draught.
3. The minimum range of stability should be:
   For L <=100 m: 20°;
   For L >=150 m: 15^0;
   For intermediate length: by interpolation.

<u>Intact stability criteria during Cargo Lifting:</u>

Recommended Design Criteria during cargo lifting as per **BV, Part D, Ch 19, sec-2.2.2.**

The following intact stability criteria are to be complied with:

- $\theta_C \leq 15°$
- $GZ_C \leq 0.6\ GZ_{MAX}$
- $A_1 \geq 0.4\ A_{tot}$



θ_C	:	Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see below Fig.)
GZ_C, GZ_{MAX}	:	Defined in above Fig.

A_1 : Area, in mrad, contained between the righting lever and the heeling arm curves, measured from the heeling angle □C to the heeling angle equal to the lesser of:

- Heeling angle θ_R of loss of stability, corresponding to the second intersection between heeling and righting arms (see below Fig.)
- Heeling angle θ_F, corresponding to flooding of unprotected openings (see below Fig.)

A_{TOT} : Total area, in mrad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b = \frac{Pd - Zz}{\Delta}$$

Where:

P : Cargo lifting weight, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane.

Z : Weight, in t, of ballast used for righting the pontoon, if applicable.

z : Transversal distance, in m, of the center of gravity of Z to the longitudinal plane.

Δ : Displacement, in t, at the loading condition considered.

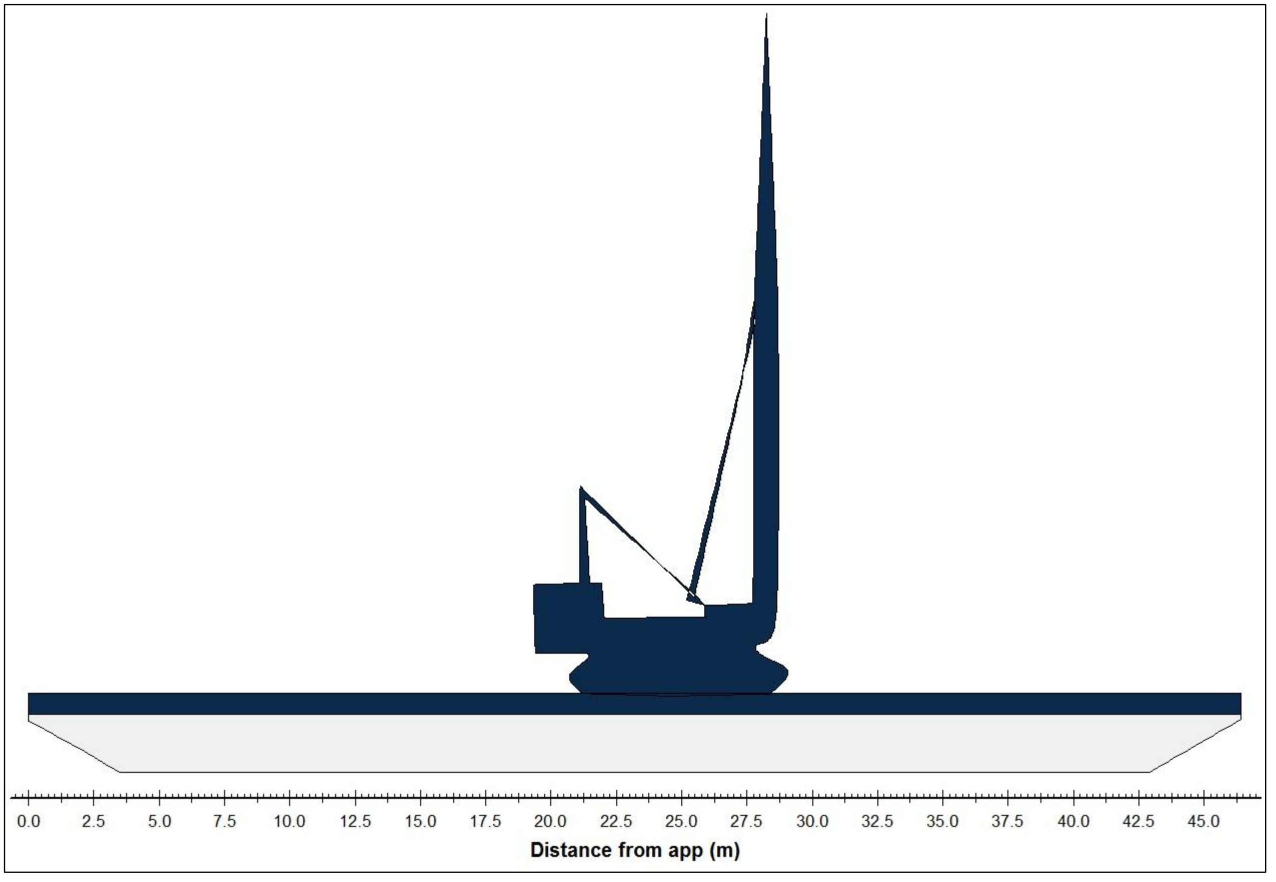
The above check is to be carried out considering the most unfavorable situations of cargo lifting combined with the lesser initial Metacentric height GM.

The vertical position of the center of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

4 DOWN FLOODING POINTS

We have considered all the openings are closed. There is no down flooding point submergence occur in any loading case. All the manholes and hatches will be watertight and closed during sailing. All air pipes are provided with necessary weather tight closing appliances.

5 WIND MOMENTS



Wind pressure:			**55.10 (kg/m^2)**					
Bilge type			**Sharp bilge**					
Draft	**Lateral area**	**LCE**	**VCE**	**Wind area**	**LCE**	**VCE**	**Windlever**	**Windmoment**
(m)	***(m^2)***	***(m)***	***(m)***	***(m^2)***	***(m)***	***(m)***	***(m)***	***(Tonnes*m)***
0.500	20.856	22.980	0.243	157.48	23.828	3.707	3.464	28.04
0.600	24.992	23.018	0.294	153.35	23.845	3.793	3.499	27.58
0.700	29.163	23.046	0.345	149.18	23.863	3.880	3.536	27.11
0.800	33.368	23.067	0.396	144.97	23.882	3.971	3.575	26.64
0.900	37.609	23.083	0.447	140.73	23.902	4.065	3.618	26.17
1.000	41.886	23.095	0.498	136.45	23.924	4.163	3.665	25.70
1.100	46.198	23.104	0.550	132.14	23.948	4.265	3.715	25.23

Wind pressure:			55.10 (kg/m^2)					
Bilge type			**Sharp bilge**					
Draft	**Lateral area**	**LCE**	**VCE**	**Wind area**	**LCE**	**VCE**	**Windlever**	**Windmoment**
(m)	***(m^2)***	***(m)***	***(m)***	***(m^2)***	***(m)***	***(m)***	***(m)***	***(Tonnes*m)***
1.200	50.545	23.112	0.601	127.79	23.973	4.370	3.769	24.76
1.300	54.928	23.118	0.653	123.41	24.001	4.481	3.828	24.28
1.400	59.346	23.123	0.705	118.99	24.032	4.598	3.893	23.81
1.500	63.800	23.127	0.757	114.54	24.065	4.720	3.963	23.33
1.600	68.289	23.130	0.809	110.05	24.101	4.849	4.040	22.85
1.700	72.814	23.133	0.861	105.52	24.141	4.986	4.125	22.37
1.800	77.374	23.135	0.914	100.96	24.185	5.133	4.219	21.89
1.900	81.969	23.136	0.966	96.368	24.234	5.289	4.323	21.41
2.000	86.598	23.138	1.019	91.740	24.287	5.458	4.439	20.93
2.100	91.238	23.141	1.071	87.100	24.345	5.639	4.568	20.45
2.200	95.877	23.144	1.124	82.460	24.409	5.836	4.712	19.97
2.300	100.52	23.147	1.176	77.821	24.481	6.049	4.874	19.50
2.400	105.16	23.149	1.227	73.181	24.563	6.284	5.057	19.02
2.500	109.80	23.151	1.279	68.541	24.655	6.543	5.264	18.55

6 TANK CAPACITIES

BALLAST

Tank description	Relative density	Volume (m^3)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)		Max FSM (Tonnes*m)
BALLAST TANK7(CS)	1.025	67.82	69.52	1.750	43.496	-3.000	(SB)	90.36
BALLAST TANK7(CP)	1.025	67.82	69.52	1.750	43.496	3.000	(PS)	90.36
BALLAST TANK1(CS)	1.025	74.89	76.77	1.726	3.120	-3.000	(SB)	97.59
BALLAST TANK1(CP)	1.025	74.89	76.77	1.726	3.120	3.000	(PS)	97.59
BALLAST TANK2(CS)	1.025	126.99	130.16	1.500	9.000	-3.000	(SB)	130.12
BALLAST TANK2(CP)	1.025	126.99	130.16	1.500	9.000	3.000	(PS)	130.12
LIUQUID TANK4(CS)	1.025	95.24	97.62	1.500	22.500	-3.000	(SB)	97.59
LIUQUID TANK4 (CP)	1.025	95.24	97.62	1.500	22.500	3.000	(PS)	97.59
BALLAST TANK5(CS)	1.025	134.04	137.39	1.500	29.000	-3.000	(SB)	137.35
BALLAST TANK5(CP)	1.025	134.04	137.39	1.500	29.000	3.000	(PS)	137.35
BALLAST TANK6(CS)	1.025	151.68	155.47	1.500	37.100	-3.000	(SB)	155.42
BALLAST TANK6(CP)	1.025	151.68	155.47	1.500	37.100	3.000	(PS)	155.42
Total		**1301.32**	**1333.85**	**1.552**	**24.566**	**0.000**	**(CL)**	**1416.84**

7 HYDROSTATICS PARTICULARS

Trim = 0.500m by aft

Draft (m)	Vol. mould (m^3)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m^2)	KMt (m)	MCT (t*m/cm)	TpCm (t/cm)
0.800	580.77	595.29	21.061	0.421	22.783	886.62	35.719	25.008	7.796
0.900	657.14	673.57	21.261	0.471	22.783	911.31	31.919	25.620	7.859
1.000	734.13	752.48	21.421	0.522	22.782	936.07	28.897	26.241	7.922
1.100	811.72	832.01	21.551	0.573	22.781	960.90	26.439	26.870	7.985
1.200	889.93	912.18	21.659	0.624	22.780	985.81	24.403	27.512	8.049
1.300	968.75	992.97	21.750	0.675	22.779	1010.79	22.690	28.164	8.112
1.400	1048.20	1074.40	21.828	0.727	22.779	1035.84	21.231	28.825	8.175
1.500	1128.24	1156.45	21.896	0.778	22.778	1060.97	19.974	29.497	8.238
1.600	1208.92	1239.14	21.954	0.830	22.777	1086.17	18.882	30.180	8.301
1.700	1290.22	1322.48	22.006	0.882	22.776	1111.45	17.924	30.872	8.364
1.800	1372.08	1406.38	22.053	0.934	22.816	1135.85	17.055	31.408	8.412
1.900	1454.30	1490.66	22.098	0.986	22.897	1159.36	16.260	31.777	8.445
2.000	1536.85	1575.27	22.143	1.038	22.979	1182.94	15.550	32.145	8.477
2.100	1619.70	1660.20	22.188	1.090	23.062	1206.59	14.914	32.510	8.509
2.200	1702.88	1745.45	22.233	1.142	23.146	1230.31	14.341	32.876	8.541
2.300	1786.32	1830.98	22.277	1.194	23.191	1253.27	13.803	33.067	8.558
2.400	1869.80	1916.55	22.318	1.245	23.194	1275.42	13.297	33.083	8.559
2.500	1953.31	2002.15	22.355	1.297	23.197	1297.57	12.838	33.095	8.560
2.600	2036.83	2087.75	22.390	1.348	23.199	1319.73	12.419	33.102	8.561
2.700	2120.34	2173.34	22.422	1.399	23.200	1341.89	12.035	33.106	8.561
2.800	2201.76	2256.81	22.471	1.449	25.520	1446.20	10.667	24.135	7.705

Note: Density of water = 1.025 tonnes/m^3

Draft, VCB, KMt, KMl is from moulded base line.

LCB and LCF values are given relative to Transom. (Fr. No-0)

Trim is taken according to LOA (=46.40 m)

Trim = 0.250 m by aft

Draft (m)	Vol. mould (m^3)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m^2)	KMt (m)	MCT (t*m/cm)	TpCm (t/cm)
0.800	579.47	593.95	22.110	0.412	22.978	885.18	35.785	24.997	7.795
0.900	655.82	672.22	22.211	0.463	22.978	909.93	31.969	25.608	7.858
1.000	732.79	751.11	22.291	0.514	22.979	934.76	28.937	26.228	7.921
1.100	810.38	830.64	22.357	0.566	22.980	959.66	26.471	26.858	7.984
1.200	888.58	910.79	22.412	0.617	22.981	984.64	24.429	27.498	8.047
1.300	967.40	991.58	22.458	0.669	22.982	1009.69	22.711	28.150	8.110
1.400	1046.82	1072.99	22.498	0.721	22.982	1034.80	21.249	28.811	8.173
1.500	1126.87	1155.04	22.532	0.772	22.983	1060.00	19.989	29.482	8.236
1.600	1207.53	1237.72	22.563	0.824	22.984	1085.28	18.894	30.164	8.299
1.700	1288.80	1321.02	22.589	0.877	22.985	1110.63	17.935	30.857	8.362
1.800	1370.70	1404.97	22.613	0.929	22.985	1136.06	17.089	31.560	8.426
1.900	1453.20	1489.53	22.634	0.981	23.006	1161.09	16.326	32.189	8.481
2.000	1536.09	1574.49	22.657	1.034	23.088	1184.78	15.612	32.568	8.514
2.100	1619.33	1659.82	22.681	1.086	23.170	1208.54	14.972	32.946	8.547
2.200	1702.79	1745.36	22.706	1.138	23.193	1231.11	14.364	33.060	8.557
2.300	1786.28	1830.93	22.729	1.190	23.195	1253.26	13.806	33.080	8.559
2.400	1869.79	1916.54	22.749	1.242	23.198	1275.41	13.299	33.092	8.560
2.500	1953.30	2002.13	22.769	1.293	23.199	1297.57	12.837	33.099	8.561
2.600	2036.83	2087.75	22.786	1.345	23.200	1319.73	12.416	33.103	8.561
2.700	2120.35	2173.36	22.802	1.396	23.200	1341.89	12.032	33.103	8.561
2.800	2203.86	2258.95	22.818	1.448	23.200	1364.05	11.680	33.102	8.561

Note: Density of water = 1.025 tonnes/m^3

Draft, VCB, KMt, KMl is from moulded base line.

LCB and LCF values are given relative to Transom. (Fr. No-0)

Trim is taken according to LOA (=46.40 m)

Trim = 0.00 m

Draft *(m)*	Vol. mould *(m^3)*	Displ. *(tonnes)*	LCB *(m)*	VCB *(m)*	LCF *(m)*	S *(m^2)*	KMt *(m)*	MCT *(t*m/cm)*	TpCm *(t/cm)*
0.800	578.96	593.43	23.162	0.408	23.172	883.98	35.815	25.000	7.795
0.900	655.31	671.69	23.163	0.460	23.174	908.80	31.993	25.610	7.858
1.000	732.29	750.59	23.164	0.511	23.177	933.70	28.956	26.231	7.921
1.100	809.87	830.12	23.166	0.563	23.179	958.67	26.487	26.860	7.984
1.200	888.07	910.27	23.167	0.615	23.181	983.72	24.442	27.502	8.047
1.300	966.89	991.06	23.168	0.667	23.184	1008.84	22.722	28.152	8.110
1.400	1046.32	1072.48	23.170	0.718	23.186	1034.03	21.258	28.813	8.173
1.500	1126.38	1154.54	23.171	0.770	23.188	1059.30	19.997	29.485	8.237
1.600	1207.05	1237.22	23.172	0.823	23.191	1084.65	18.901	30.167	8.300
1.700	1288.31	1320.52	23.173	0.875	23.193	1110.08	17.941	30.859	8.363
1.800	1370.21	1404.47	23.175	0.927	23.195	1135.58	17.094	31.562	8.426
1.900	1452.73	1489.04	23.176	0.979	23.198	1161.16	16.342	32.276	8.489
2.000	1535.85	1574.25	23.177	1.032	23.200	1186.81	15.671	33.001	8.552
2.100	1619.29	1659.77	23.178	1.084	23.200	1208.95	14.984	33.037	8.555
2.200	1702.78	1745.35	23.179	1.137	23.200	1231.10	14.365	33.065	8.558
2.300	1786.27	1830.93	23.180	1.189	23.200	1253.25	13.807	33.084	8.559
2.400	1869.78	1916.53	23.181	1.241	23.200	1275.41	13.300	33.097	8.560
2.500	1953.31	2002.14	23.182	1.292	23.200	1297.57	12.837	33.102	8.561
2.600	2036.82	2087.74	23.183	1.344	23.200	1319.73	12.415	33.101	8.561
2.700	2120.34	2173.35	23.183	1.395	23.200	1341.89	12.031	33.101	8.561
2.800	2203.86	2258.96	23.184	1.447	23.200	1364.05	11.679	33.101	8.561

Note: Density of water = 1.025 tonnes/m^3

Draft, VCB, KMt, KMl is from moulded base line.

LCB and LCF values are given relative to Transom. (Fr. No-0)

Trim is taken according to LOA (=46.40 m)

Trim = 0.250 m by Bow

Draft (m)	Vol. mould (m^3)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m^2)	KMt (m)	MCT (t*m/cm)	TpCm (t/cm)
0.800	579.24	593.72	24.215	0.411	23.366	883.04	35.810	25.018	7.797
0.900	655.61	672.00	24.116	0.463	23.370	907.93	31.990	25.629	7.860
1.000	732.60	750.91	24.038	0.514	23.374	932.90	28.954	26.250	7.923
1.100	810.20	830.46	23.974	0.565	23.378	957.94	26.485	26.880	7.986
1.200	888.42	910.63	23.922	0.617	23.382	983.06	24.441	27.522	8.049
1.300	967.25	991.43	23.878	0.669	23.386	1008.26	22.722	28.171	8.112
1.400	1046.71	1072.88	23.841	0.720	23.390	1033.52	21.258	28.834	8.175
1.500	1126.76	1154.93	23.809	0.772	23.394	1058.87	19.997	29.504	8.238
1.600	1207.45	1237.63	23.782	0.824	23.398	1084.30	18.902	30.186	8.301
1.700	1288.75	1320.97	23.757	0.877	23.402	1109.80	17.942	30.880	8.364
1.800	1370.66	1404.92	23.736	0.929	23.406	1135.38	17.095	31.583	8.428
1.900	1453.18	1489.51	23.718	0.981	23.389	1160.60	16.331	32.210	8.483
2.000	1536.09	1574.49	23.698	1.034	23.309	1184.55	15.614	32.580	8.515
2.100	1619.33	1659.81	23.676	1.086	23.229	1208.51	14.972	32.948	8.547
2.200	1702.79	1745.36	23.653	1.138	23.207	1231.11	14.365	33.060	8.557
2.300	1786.29	1830.95	23.632	1.190	23.205	1253.26	13.806	33.080	8.559
2.400	1869.79	1916.54	23.613	1.242	23.202	1275.41	13.299	33.092	8.560
2.500	1953.30	2002.13	23.595	1.293	23.201	1297.57	12.837	33.099	8.561
2.600	2036.82	2087.74	23.579	1.345	23.200	1319.73	12.416	33.103	8.561
2.700	2120.34	2173.35	23.564	1.396	23.200	1341.89	12.032	33.102	8.561
2.800	2203.88	2258.98	23.550	1.448	23.200	1364.05	11.680	33.102	8.561

Note: Density of water = 1.025 tonnes/m^3

Draft, VCB, KMt, KMl is from moulded base line.

LCB and LCF values are given relative to Transom. (Fr. No-0)

Trim is taken according to LOA (=46.40 m)

Trim = 0.500 m by Bow

Draft (m)	Vol. mould (m^3)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m^2)	KMt (m)	MCT (t*m/cm)	TpCm (t/cm)
0.800	580.32	594.83	25.265	0.421	23.560	882.34	35.770	25.052	7.800
0.900	656.72	673.14	25.067	0.471	23.566	907.31	31.960	25.663	7.863
1.000	733.73	752.08	24.910	0.522	23.572	932.35	28.931	26.284	7.926
1.100	811.36	831.65	24.782	0.573	23.577	957.47	26.467	26.916	7.989
1.200	889.62	911.86	24.676	0.624	23.583	982.67	24.427	27.557	8.052
1.300	968.47	992.69	24.588	0.675	23.588	1007.95	22.711	28.209	8.116
1.400	1047.95	1074.15	24.512	0.727	23.594	1033.29	21.249	28.872	8.179
1.500	1128.05	1156.25	24.447	0.778	23.600	1058.72	19.990	29.542	8.242
1.600	1208.76	1238.98	24.391	0.830	23.605	1084.22	18.896	30.225	8.305
1.700	1290.09	1322.34	24.341	0.882	23.611	1109.81	17.937	30.919	8.368
1.800	1372.00	1406.30	24.297	0.934	23.574	1134.57	17.065	31.451	8.416
1.900	1454.25	1490.60	24.254	0.986	23.495	1158.47	16.267	31.811	8.448
2.000	1536.82	1575.24	24.211	1.038	23.416	1182.38	15.555	32.167	8.479
2.100	1619.70	1660.19	24.168	1.090	23.335	1206.30	14.917	32.525	8.511
2.200	1702.87	1745.45	24.126	1.142	23.253	1230.23	14.342	32.880	8.542
2.300	1786.32	1830.98	24.083	1.194	23.209	1253.27	13.803	33.067	8.558
2.400	1869.81	1916.56	24.044	1.245	23.206	1275.42	13.297	33.083	8.559
2.500	1953.32	2002.15	24.008	1.297	23.203	1297.57	12.838	33.095	8.560
2.600	2036.82	2087.74	23.975	1.348	23.201	1319.73	12.419	33.102	8.561
2.700	2120.34	2173.34	23.945	1.399	23.200	1341.89	12.035	33.106	8.561
2.800	2201.78	2256.83	23.897	1.449	20.880	1446.20	10.667	24.135	7.705

Note: Density of water = 1.025 tonnes/m^3

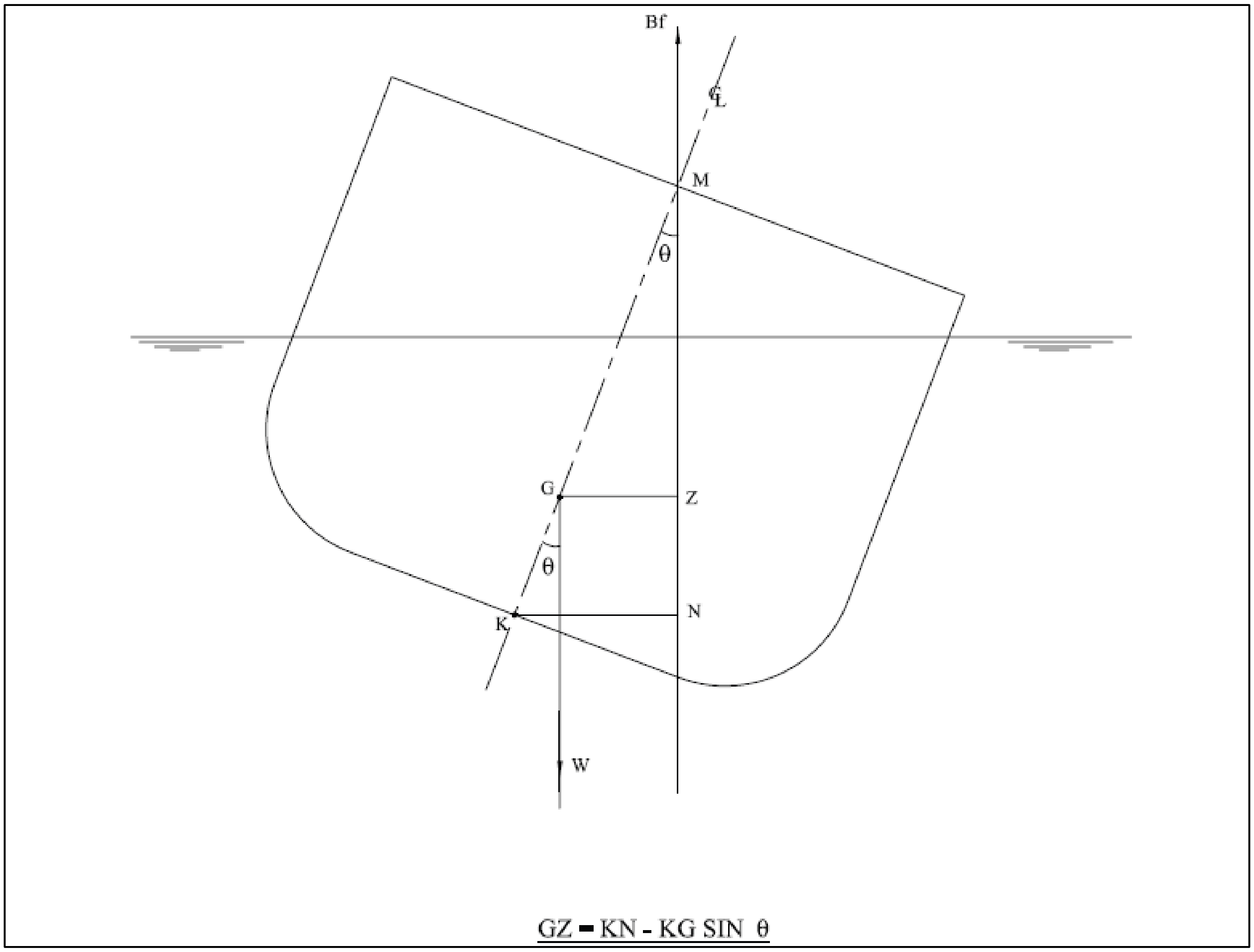
Draft, VCB, KMt, KMl is from moulded base line.

LCB and LCF values are given relative to Transom. (Fr. No-0)

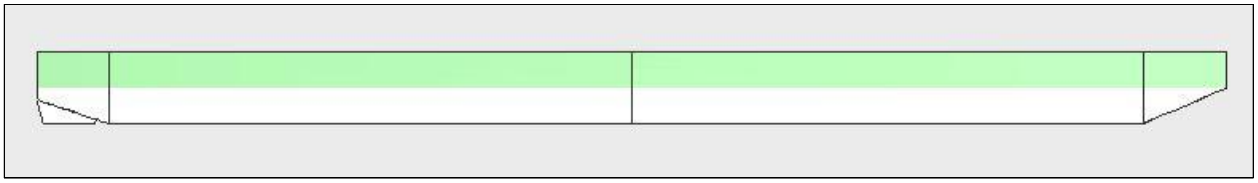
Trim is taken according to LOA (=46.40 m)

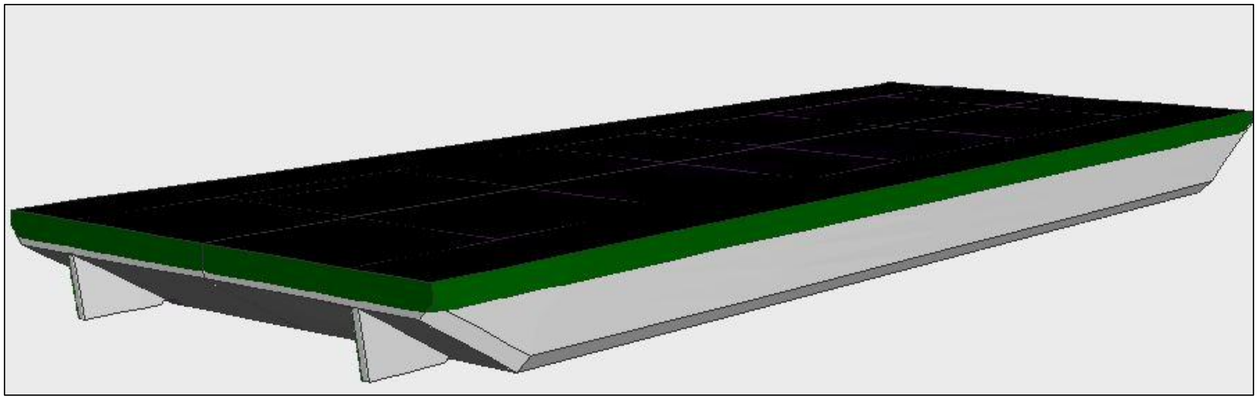
8 CROSS CURVES OF STABILITY

8.1 KN CONCEPT DIAGRAM



Volume considered for calculation of KN is depicted below:





8.2 KN VALUES

KN Values for Initial trim: 0.500 m by aft:

Displ.	0.0บ	2.0บ	5.0บ	10.0บ	15.0บ	20.0บ	30.0บ	40.0บ	50.0บ	60.0บ
400.00	0.000	1.769	3.966	5.510	6.164	6.502	6.653	6.342	5.730	4.899
450.00	0.000	1.595	3.701	5.333	6.030	6.395	6.553	6.240	5.638	4.825
500.00	0.000	1.453	3.453	5.164	5.902	6.290	6.444	6.131	5.541	4.747
550.00	0.000	1.335	3.225	5.002	5.781	6.181	6.327	6.016	5.440	4.666
600.00	0.000	1.236	3.017	4.847	5.663	6.069	6.203	5.897	5.336	4.583
650.00	0.000	1.150	2.829	4.699	5.548	5.952	6.073	5.774	5.229	4.499
700.00	0.000	1.076	2.660	4.555	5.433	5.829	5.939	5.648	5.121	4.414
750.00	0.000	1.012	2.510	4.415	5.316	5.701	5.801	5.521	5.012	4.328
800.00	0.000	0.956	2.378	4.280	5.197	5.567	5.661	5.391	4.901	4.241
850.00	0.000	0.906	2.259	4.149	5.073	5.427	5.518	5.261	4.790	4.153
900.00	0.000	0.862	2.153	4.023	4.946	5.285	5.374	5.129	4.677	4.065
950.00	0.000	0.823	2.057	3.899	4.815	5.139	5.228	4.996	4.564	3.976
1000.00	0.000	0.788	1.970	3.777	4.678	4.990	5.080	4.862	4.451	3.887
1050.00	0.000	0.756	1.891	3.656	4.537	4.839	4.931	4.727	4.337	3.798
1100.00	0.000	0.727	1.818	3.536	4.393	4.686	4.782	4.592	4.222	3.709
1150.00	0.000	0.701	1.752	3.416	4.245	4.531	4.631	4.457	4.108	3.619
1200.00	0.000	0.677	1.691	3.297	4.095	4.374	4.480	4.321	3.993	3.529

KN Values for Initial trim: 0.250 m by aft

Displ.	0.0บ	2.0บ	5.0บ	10.0บ	15.0บ	20.0บ	30.0บ	40.0บ	50.0บ	60.0บ
400.00	0.000	1.774	4.034	5.549	6.193	6.525	6.701	6.391	5.773	4.935
450.00	0.000	1.599	3.758	5.367	6.055	6.419	6.602	6.286	5.679	4.858
500.00	0.000	1.457	3.497	5.194	5.925	6.318	6.491	6.174	5.579	4.778
550.00	0.000	1.338	3.252	5.029	5.801	6.215	6.371	6.057	5.476	4.696
600.00	0.000	1.237	3.030	4.872	5.683	6.108	6.245	5.936	5.371	4.613
650.00	0.000	1.151	2.834	4.721	5.569	5.993	6.113	5.811	5.263	4.527
700.00	0.000	1.077	2.662	4.576	5.458	5.869	5.977	5.685	5.154	4.441
750.00	0.000	1.012	2.512	4.436	5.346	5.740	5.839	5.556	5.044	4.354
800.00	0.000	0.956	2.380	4.301	5.230	5.604	5.697	5.426	4.932	4.267
850.00	0.000	0.906	2.261	4.169	5.109	5.464	5.554	5.294	4.820	4.179
900.00	0.000	0.862	2.155	4.040	4.983	5.320	5.408	5.161	4.707	4.090
950.00	0.000	0.823	2.059	3.916	4.849	5.173	5.261	5.028	4.594	4.001
1000.00	0.000	0.787	1.971	3.794	4.712	5.024	5.113	4.894	4.480	3.912
1050.00	0.000	0.756	1.892	3.675	4.570	4.872	4.964	4.759	4.365	3.822
1100.00	0.000	0.727	1.819	3.557	4.426	4.718	4.814	4.623	4.251	3.733
1150.00	0.000	0.700	1.753	3.439	4.278	4.563	4.663	4.487	4.135	3.643
1200.00	0.000	0.676	1.692	3.320	4.128	4.406	4.511	4.351	4.020	3.553

KN Values for Initial trim: 0.00 m

Displ.	0.0บ	2.0บ	5.0บ	10.0บ	15.0บ	20.0บ	30.0บ	40.0บ	50.0บ	60.0บ
400.00	0.000	1.776	4.057	5.563	6.202	6.533	6.718	6.407	5.788	4.947
450.00	0.000	1.601	3.778	5.378	6.064	6.426	6.618	6.301	5.692	4.869
500.00	0.000	1.459	3.514	5.203	5.933	6.325	6.506	6.189	5.592	4.789
550.00	0.000	1.340	3.263	5.038	5.808	6.228	6.386	6.071	5.489	4.707
600.00	0.000	1.238	3.033	4.880	5.689	6.122	6.259	5.949	5.383	4.622
650.00	0.000	1.151	2.835	4.728	5.575	6.007	6.127	5.824	5.275	4.537
700.00	0.000	1.076	2.663	4.583	5.465	5.883	5.990	5.697	5.165	4.451
750.00	0.000	1.012	2.513	4.443	5.357	5.752	5.851	5.568	5.055	4.364
800.00	0.000	0.956	2.381	4.307	5.243	5.616	5.709	5.437	4.943	4.276
850.00	0.000	0.906	2.262	4.175	5.122	5.476	5.566	5.306	4.831	4.188
900.00	0.000	0.862	2.156	4.047	4.994	5.332	5.420	5.173	4.718	4.099
950.00	0.000	0.823	2.059	3.922	4.861	5.185	5.273	5.039	4.604	4.010
1000.00	0.000	0.787	1.971	3.800	4.723	5.035	5.124	4.904	4.489	3.920
1050.00	0.000	0.756	1.892	3.681	4.582	4.883	4.975	4.768	4.376	3.831
1100.00	0.000	0.727	1.820	3.565	4.437	4.730	4.825	4.634	4.260	3.742
1150.00	0.000	0.701	1.753	3.449	4.289	4.574	4.673	4.497	4.145	3.651
1200.00	0.000	0.677	1.693	3.329	4.139	4.417	4.522	4.361	4.029	3.561

KN Values for Initial trim: 0.250 m by bow

Displ.	0.0บ	2.0บ	5.0บ	10.0บ	15.0บ	20.0บ	30.0บ	40.0บ	50.0บ	60.0บ
400.00	0.000	1.774	4.035	5.549	6.192	6.525	6.701	6.391	5.773	4.935
450.00	0.000	1.600	3.759	5.367	6.055	6.419	6.602	6.286	5.679	4.858
500.00	0.000	1.458	3.498	5.194	5.925	6.318	6.491	6.174	5.579	4.779
550.00	0.000	1.339	3.254	5.029	5.802	6.216	6.372	6.057	5.477	4.697
600.00	0.000	1.238	3.032	4.872	5.683	6.108	6.245	5.936	5.371	4.613
650.00	0.000	1.152	2.835	4.721	5.569	5.993	6.114	5.812	5.264	4.528
700.00	0.000	1.077	2.663	4.576	5.458	5.870	5.978	5.685	5.155	4.442
750.00	0.000	1.012	2.513	4.437	5.346	5.740	5.839	5.557	5.045	4.355
800.00	0.000	0.956	2.381	4.301	5.231	5.604	5.698	5.427	4.933	4.268
850.00	0.000	0.906	2.262	4.170	5.110	5.464	5.555	5.295	4.821	4.179
900.00	0.000	0.862	2.156	4.042	4.983	5.321	5.409	5.162	4.708	4.091
950.00	0.000	0.823	2.059	3.917	4.850	5.174	5.262	5.029	4.595	4.002
1000.00	0.000	0.788	1.972	3.796	4.713	5.025	5.114	4.895	4.481	3.913
1050.00	0.000	0.756	1.893	3.677	4.571	4.873	4.965	4.760	4.365	3.823
1100.00	0.000	0.727	1.820	3.559	4.427	4.719	4.815	4.624	4.251	3.733
1150.00	0.000	0.701	1.753	3.441	4.279	4.564	4.664	4.488	4.136	3.643
1200.00	0.000	0.677	1.693	3.322	4.129	4.407	4.512	4.352	4.021	3.553

KN Values for Initial trim: 0.500 m by bow

Displ.	0.0ບ	2.0ບ	5.0ບ	10.0ບ	15.0ບ	20.0ບ	30.0ບ	40.0ບ	50.0ບ	60.0ບ
400.00	0.000	1.770	3.967	5.509	6.164	6.502	6.654	6.343	5.732	4.902
450.00	0.000	1.596	3.703	5.333	6.030	6.396	6.554	6.242	5.640	4.827
500.00	0.000	1.455	3.456	5.164	5.903	6.290	6.446	6.133	5.542	4.748
550.00	0.000	1.336	3.228	5.002	5.781	6.182	6.329	6.017	5.441	4.667
600.00	0.000	1.237	3.020	4.848	5.664	6.071	6.205	5.898	5.337	4.584
650.00	0.000	1.151	2.831	4.699	5.549	5.954	6.075	5.776	5.231	4.500
700.00	0.000	1.077	2.662	4.556	5.434	5.831	5.941	5.650	5.123	4.415
750.00	0.000	1.013	2.512	4.417	5.318	5.703	5.803	5.523	5.014	4.329
800.00	0.000	0.957	2.380	4.282	5.198	5.569	5.663	5.394	4.903	4.242
850.00	0.000	0.907	2.261	4.152	5.075	5.430	5.521	5.263	4.792	4.155
900.00	0.000	0.863	2.155	4.026	4.948	5.287	5.376	5.131	4.680	4.067
950.00	0.000	0.823	2.059	3.902	4.817	5.141	5.230	4.998	4.567	3.978
1000.00	0.000	0.788	1.971	3.780	4.680	4.992	5.083	4.865	4.453	3.889
1050.00	0.000	0.756	1.893	3.659	4.539	4.841	4.934	4.730	4.339	3.800
1100.00	0.000	0.727	1.820	3.539	4.395	4.688	4.784	4.595	4.225	3.711
1150.00	0.000	0.701	1.753	3.419	4.248	4.533	4.634	4.460	4.110	3.621
1200.00	0.000	0.677	1.692	3.300	4.099	4.377	4.483	4.323	3.995	3.531

9 CRANE DETAILS

Crane with self-weight of 218 MT(including counter weight and boom) is operated on the deck having a boom length 40.0 m having working radius of 16 m with 45 MT capacity .the stability is checked when the crane is at stowage location and in operating condition.

- VCG of the crane is considered 3.00 m above the deck level.

10 LIGHTSHIP PARTICULARS

The estimated lightship particulars estimated from the weight estimation and the values are given below. The VCG is considered at main deck level. The final lightship values will get after the draft survey.

Lightship weight	=	500.00MT
VCG	=	3.00m above base line
LCG	=	23.200 m from Transom
TCG	=	0.00 m

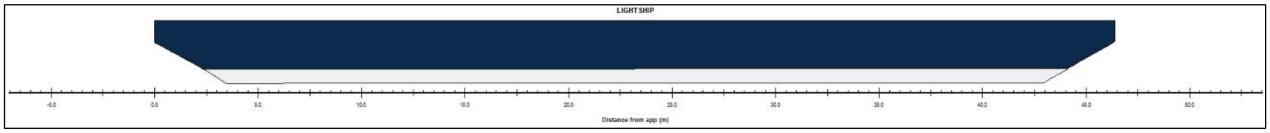
11 LOADING CONDITIONS

The following are the loading conditions considered in the booklet and the stability of the“46.4 M FLAT TOP PONTOON “these conditions is checked as per the criteria mentioned in Section 3.

1. LC-01- Lightship condition
2. LC-02-fully Loading condition without crane hook load
3. LC-03-Loading condition with crane at center operating towards port side
4. LC-04-Loading condition with crane at center operating towards port side with 45 degree
5. LC-05- Loading condition with crane at Aft operating towards forward

11.1 LC-01- LIGHTSHIP CONDITION

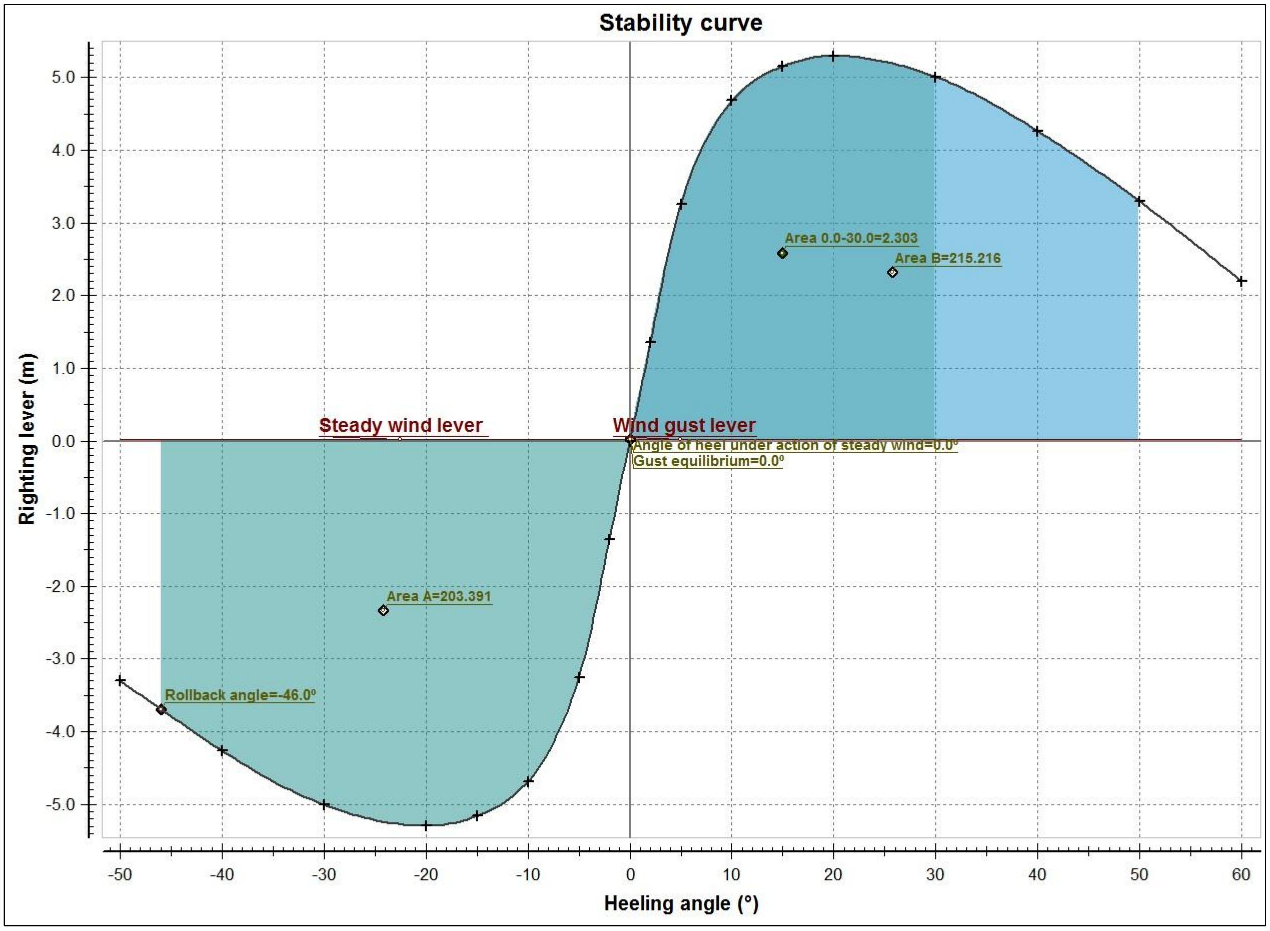
The present load case for the intact stability analysis has been done by considering the Lightship weight of the pontoon



Description	Density *(t/m^3)*	Fill%	Weight *(tonnes)*	VCG *(m)*	LCG *(m)*	TCG *(m)*	FSM *(Tonnes*m)*
Lightship			500.00	3.000	23.200	0.000 (CL)	
Deadweight			0.00	0.000	0.000	0.000 (CL)	0.00
Displacement			500.00	3.000	23.200	0.000 (CL)	0.00

Hydrostatic particulars			
List	0.00 (CL) *(Degr.)*	**KM**	41.980 *(m)*
Draft aft pp	0.675 *(m)*	**GM solid**	38.980 *(m)*
Mean draft	0.679 *(m)*	**GG'**	0.000 *(m)*
Draft forward pp	0.683 *(m)*	**GM liquid**	38.980 *(m)*
Trim	0.008 *(m)*	**MCT**	24.276 *(t*m/cm)*
Volume	487.57 *(m^3)*	**Relative water density**	1.0250
LCF	23.175 *(m)*	**Immersion rate**	7.719 *(t/cm)*

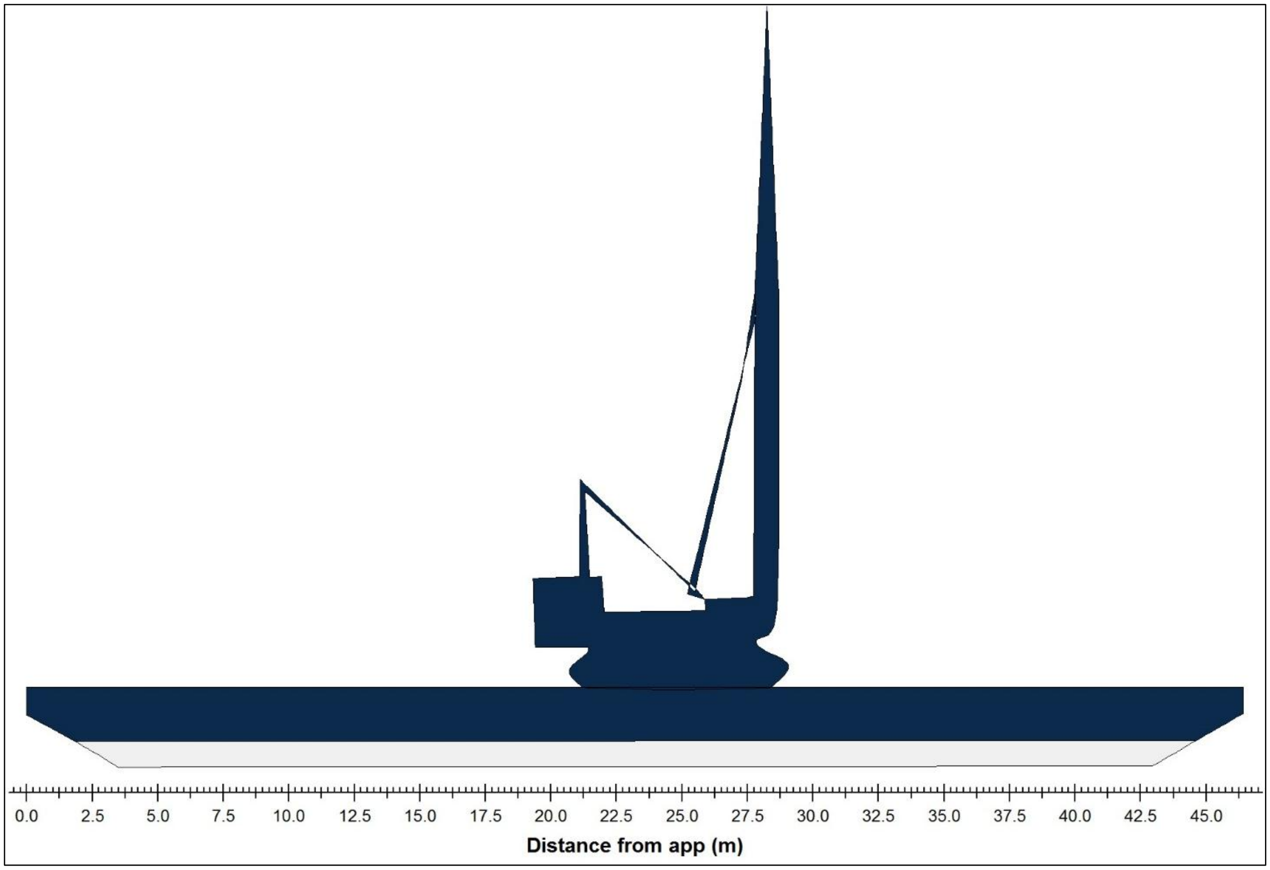
GZ values	
Heeling angle *(Degr.)*	**GZ** *(m)*
0.00 (CL)	0.000
2.00 (PS)	1.354
5.00 (PS)	3.253
10.00 (PS)	4.682
15.00 (PS)	5.157
20.00 (PS)	5.299
30.00 (PS)	5.006
40.00 (PS)	4.260
50.00 (PS)	3.294
60.00 (PS)	2.191



Evaluation of criteria				
Minimum design criteria applicable to all ships				
Description	**Attained value**	**Criterion**	**Required value**	**Complies**
Area 0º - 30º	2.3025 *(mrad)*	>=	0.0800 *(mrad)*	YES
Severe wind and rolling criterion				YES
Wind pressure	55.1 *(kg/m^2)*			
Wind area	104.70 *(m^2)*			
Steady wind lever	0.016 *(m)*			
Deck immersion angle	21.27 (Degr.)			
Maximum allowed static heeling angle	0.0 *(Degr.)*	<=	16.0 *(Degr.)*	YES
Max allowed ratio static angle/deck immersion angle	0.001	<=	0.500	YES
Range of stability	60.0 *(Degr.)*	>=	20.0 *(Degr.)*	YES
The vessel complies with the stability criteria				

11.2 LC-02-FULLY LOADING CONDITION FOR DESIGNED DRAFT

The present load case for the intact stability analysis has been done by considering the Lightship weight of the pontoon, and other deck equipment's without hook load and misc. items stowed on deck

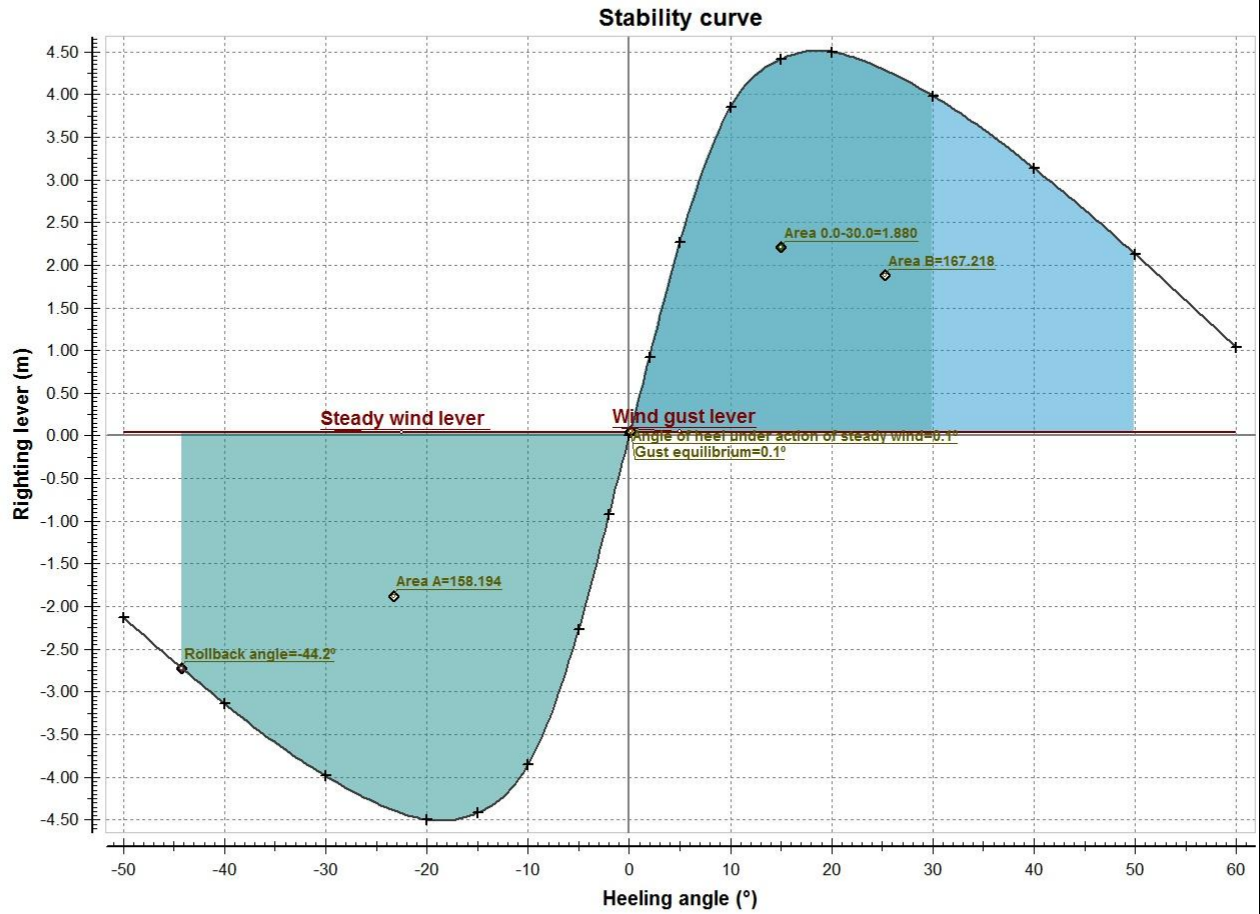


Description	**Density** *(t/m^3)*	**Fill%**	**Weight** *(tonnes)*	**VCG** *(m)*	**LCG** *(m)*	**TCG** *(m)*	**FSM** *(Tonnes*m)*
Miscellaneous							
CRANE			218.00	6.000	23.200	0.000 (CL)	
Totals for Miscellaneous			**218.00**	**6.000**	**23.200**	**0.000 (CL)**	
Lightship			500.00	3.000	23.200	0.000 (CL)	
Deadweight			218.00	6.000	23.200	0.000 (CL)	0.00
Displacement			718.00	3.911	23.200	0.000 (CL)	0.00

Hydrostatic particulars					
List	0.00 (CL)	*(Degr.)*	**KM**	30.136	*(m)*
Draft aft pp	0.953	*(m)*	**GM solid**	26.225	*(m)*
Mean draft	0.959	*(m)*	**GG'**	0.000	*(m)*
Draft forward pp	0.964	*(m)*	**GM liquid**	26.225	*(m)*
Trim	0.010	*(m)*	**MCT**	25.972	*(t*m/cm)*
Volume	700.31	*(m^3)*	**Relative water density**	1.0250	
LCF	23.184	*(m)*	**Immersion rate**	7.895	*(t/cm)*

GZ values

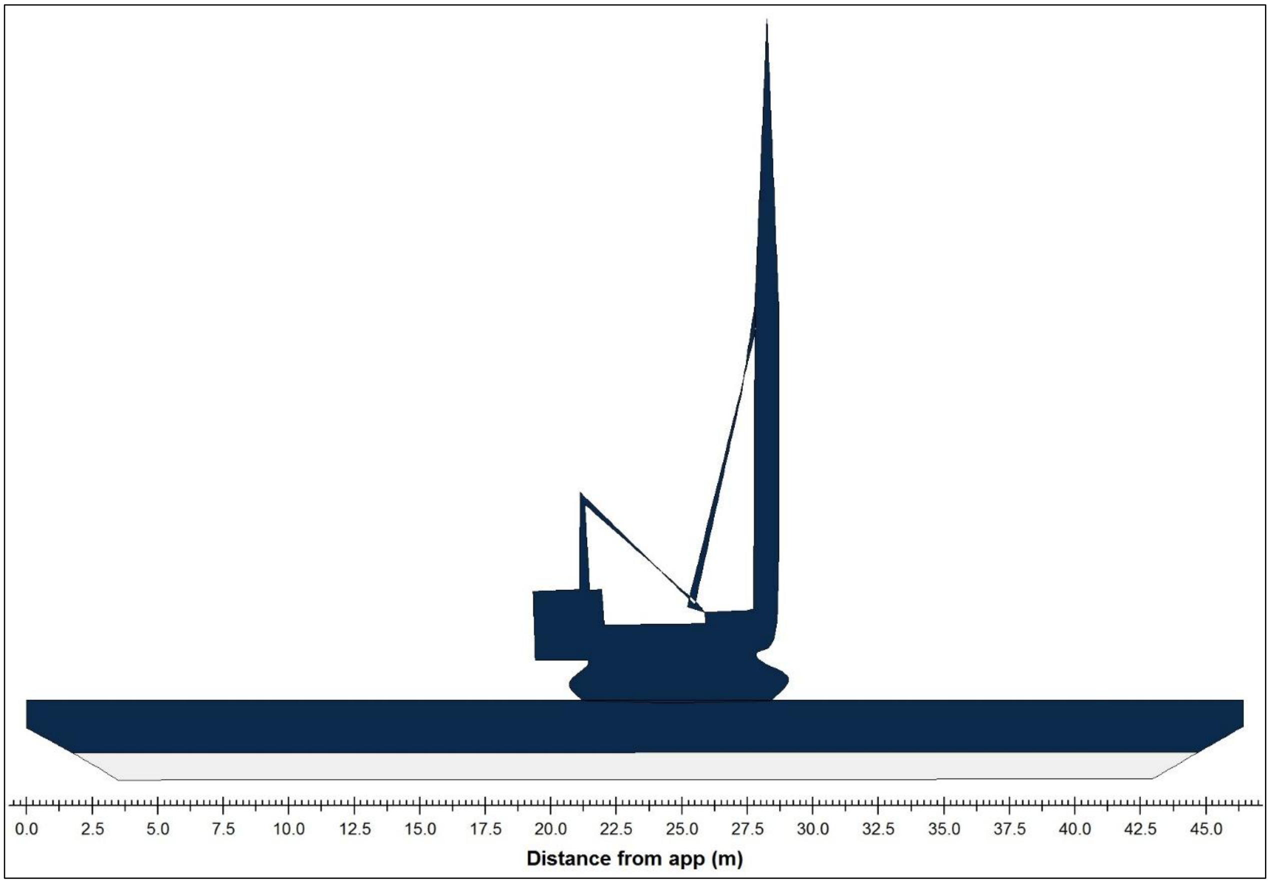
Heeling angle *(Degr.)*	GZ *(m)*
0.00 (CL)	0.000
2.00 (PS)	0.916
5.00 (PS)	2.266
10.00 (PS)	3.853
15.00 (PS)	4.413
20.00 (PS)	4.499
30.00 (PS)	3.985
40.00 (PS)	3.137
50.00 (PS)	2.130
60.00 (PS)	1.033



Evaluation of criteria				
Minimum design criteria applicable to all ships				
Description	**Attained value**	**Criterion**	**Required value**	**Complies**
Area 0º - 30º	1.8796 *(mrad)*	>=	0.0800 *(mrad)*	YES
Severe wind and rolling criterion				YES
Wind pressure	55.1 *(kg/m^2)*			
Wind area	138.23 *(m^2)*			
Steady wind lever	0.036 *(m)*			
Deck immersion angle	15.22 (Degr.)			
Maximum allowed static heeling angle	0.1 *(Degr.)*	<=	16.0 *(Degr.)*	YES
Max allowed ratio static angle/deck immersion angle	0.005	<=	0.500	YES
Range of stability	60.0 *(Degr.)*	>=	20.0 *(Degr.)*	YES
The vessel complies with the stability criteria				

11.3 LC-03- LOADING CONDITION WITH CRANE LOCATED AT CENTER AND HOOK LOAD TOWARDS PORT SIDE

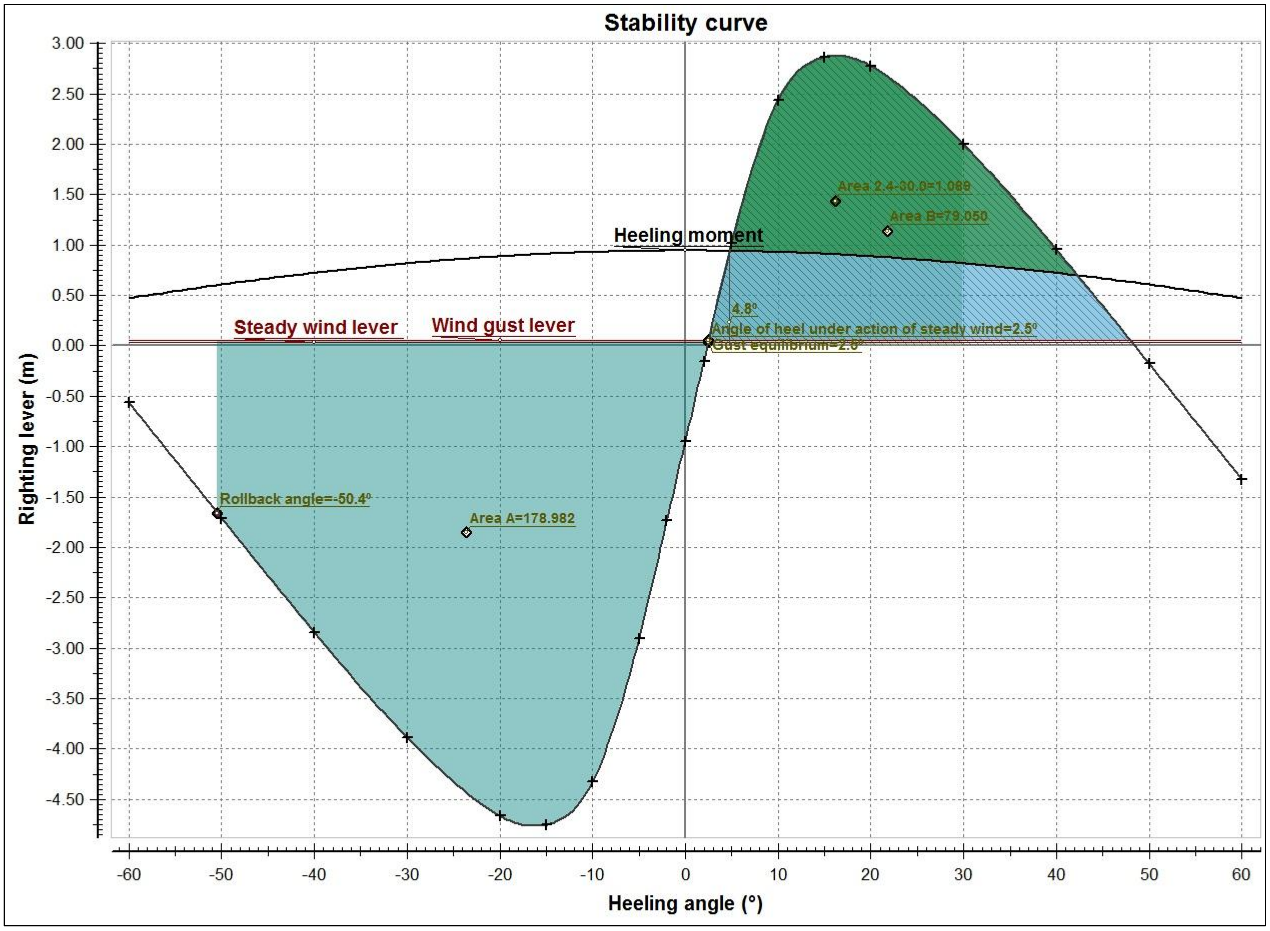
The present load case for the intact stability analysis has been done by considering the Lightship weight of the pontoon. Crane is located at the center and operated towards port side and it is operating with a radius of 16.0m towards port side from the crane location. Total moment of 720 t.m is applied .It is not recommended to operate the crane with more than the mentioned operating radius for the particular load.



Description	Density (t/m^3)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Miscellaneous							
CRANE			218.00	6.000	23.200	0.000 (CL)	
HOOK LOAD			45.00	39.327	23.200	16.000 (PS)	
Totals for Miscellaneous			**263.00**	**11.702**	**23.200**	**2.738 (PS)**	
Lightship			500.00	3.000	23.200	0.000 (CL)	
Deadweight			263.00	11.702	23.200	2.738 (PS)	0.00
Displacement			763.00	6.000	23.200	0.944 (PS)	0.00

Hydrostatic particulars			
List	2.39 (PS) *(Degr.)*	**KM**	28.542 *(m)*
Draft aft pp	1.008 *(m)*	**GM solid**	22.543 *(m)*
Mean draft	1.014 *(m)*	**GG'**	0.000 *(m)*
Draft forward pp	1.019 *(m)*	**GM liquid**	22.543 *(m)*
Trim	0.010 *(m)*	**MCT**	26.363 *(t*m/cm)*
Volume	744.25 *(m^3)*	**Relative water density**	1.0250
LCF	23.185 *(m)*	**Immersion rate**	7.937 *(t/cm)*

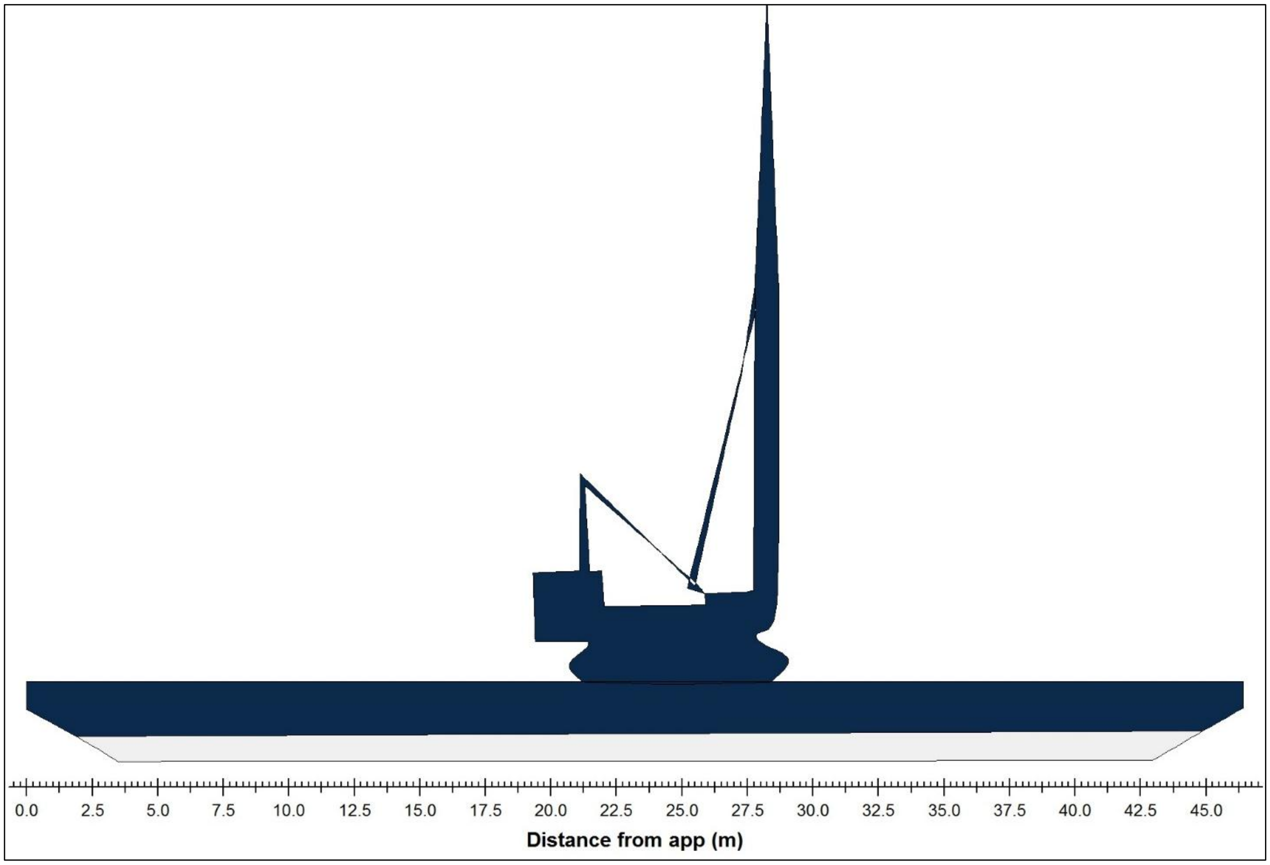
GZ values	
Heeling angle *(Degr.)*	**GZ *(m)***
0.00 (CL)	-0.944
2.00 (PS)	-0.156
5.00 (PS)	1.014
10.00 (PS)	2.436
15.00 (PS)	2.864
20.00 (PS)	2.778
30.00 (PS)	1.998
40.00 (PS)	0.955
50.00 (PS)	-0.177
60.00 (PS)	-1.327


Stability curve
Righting lever (m)
Heeling angle (°)
3.00
2.50
2.00
1.50
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-3.50
-4.00
-4.50
-60
-50
-40
-30
-20
-10
0
10
20
30
40
50
60
Heeling moment
Steady wind lever
Wind gust lever
Area 2.4-30.0=1.089
Area B=79.050
4.8°
Angle of heel under action of steady wind=2.5°
Gust equilibrium=2.6°
Rollback angle=-50.4°
Area A=178.982

Evaluation of criteria				
Minimum design criteria applicable to all ships				
Description	**Attained value**	**Criterion**	**Required value**	**Complies**
Area 0º - 30º	1.0892 *(mrad)*	>=	0.0800 *(mrad)*	**YES**
Severe wind and rolling criterion				**YES**
Wind pressure	55.1 *(kg/m^2)*			
Wind area	135.87 *(m^2)*			
Steady wind lever	0.034 *(m)*			
Deck immersion angle	14.42 (Degr.)			
Maximum allowed static heeling angle	2.5 *(Degr.)*	<=	16.0 *(Degr.)*	**YES**
Max allowed ratio static angle/deck immersion angle	0.171	<=	0.500	**YES**
Range of stability	46.1 *(Degr.)*	>=	20.0 *(Degr.)*	**YES**
Crane criteria				
Heeling moment	4.8 *(Degr.)*	<	15.0 *(Degr.)*	**YES**
Heeling moment	720.00 *(Tonnes*m)*			
Attained value smaller than deck immersion angle	4.8 *(Degr.)*	<	14.4 *(Degr.)*	**YES**
ratio of area				**YES**
Heeling moment:	Heeling moment			
Lower angle	4.8 *(Degr.)*			
Upper angle	42.3 *(Degr.)*			
Area A	0.8062 *(mrad)*			
Area B	1.4201 *(mrad)*			
Ratio of areaA/areaB	0.568	>=	0.400	**YES**
GZc/GZmax	0.327	<=	0.600	**YES**
The vessel complies with the stability criteria				

11.4 LC-03- LOADING CONDITION WITH CRANE LOCATED AT CENTER AND HOOK LOAD TOWARDS PORT SIDE WITH 45 DEGREES

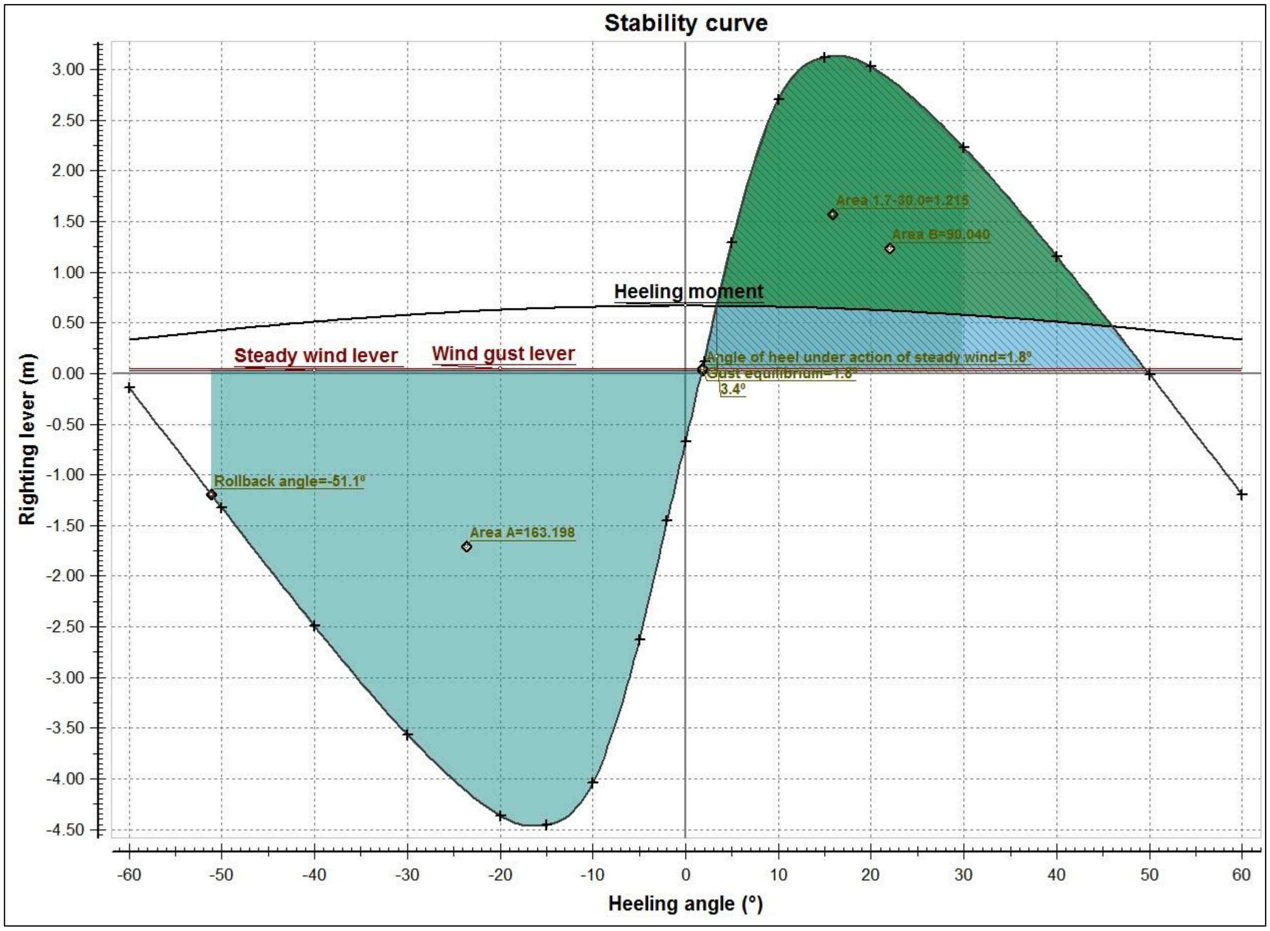
The present load case for the intact stability analysis has been done by considering the Lightship weight of the pontoon. Crane is located at the center and operated towards port side and it is operating with a radius of 11.308m towards port side with 45 degrees angle from the crane location. It is not recommended to operate the crane with more than the mentioned operating radius for the particular load.



Description	Density (t/m^3)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Miscellaneous							
CRANE			218.00	6.000	23.200	0.000 (CL)	
HOOK LOAD			45.00	39.327	34.519	11.308 (PS)	
Totals for Miscellaneous			**263.00**	**11.702**	**25.137**	**1.935 (PS)**	
Lightship			500.00	3.000	23.200	0.000 (CL)	
Deadweight			263.00	11.702	25.137	1.935 (PS)	0.00
Displacement			763.00	6.000	23.868	0.667 (PS)	0.00

Hydrostatic particulars			
List	1.69 (PS) *(Degr.)*	**KM**	28.543 *(m)*
Draft aft pp	0.909 *(m)*	**GM solid**	22.544 *(m)*
Mean draft	1.014 *(m)*	**GG'**	0.000 *(m)*
Draft forward pp	1.120 *(m)*	**GM liquid**	22.544 *(m)*
Trim	0.211 *(m)*	**MCT**	26.358 *(t*m/cm)*
Volume	744.27 *(m^3)*	**Relative water density**	1.0250
LCF	23.343 *(m)*	**Immersion rate**	7.935 *(t/cm)*

GZ values	
Heeling angle *(Degr.)*	**GZ** *(m)*
0.00 (CL)	-0.667
2.00 (PS)	0.121
5.00 (PS)	1.290
10.00 (PS)	2.704
15.00 (PS)	3.123
20.00 (PS)	3.030
30.00 (PS)	2.228
40.00 (PS)	1.158
50.00 (PS)	-0.007
60.00 (PS)	-1.195

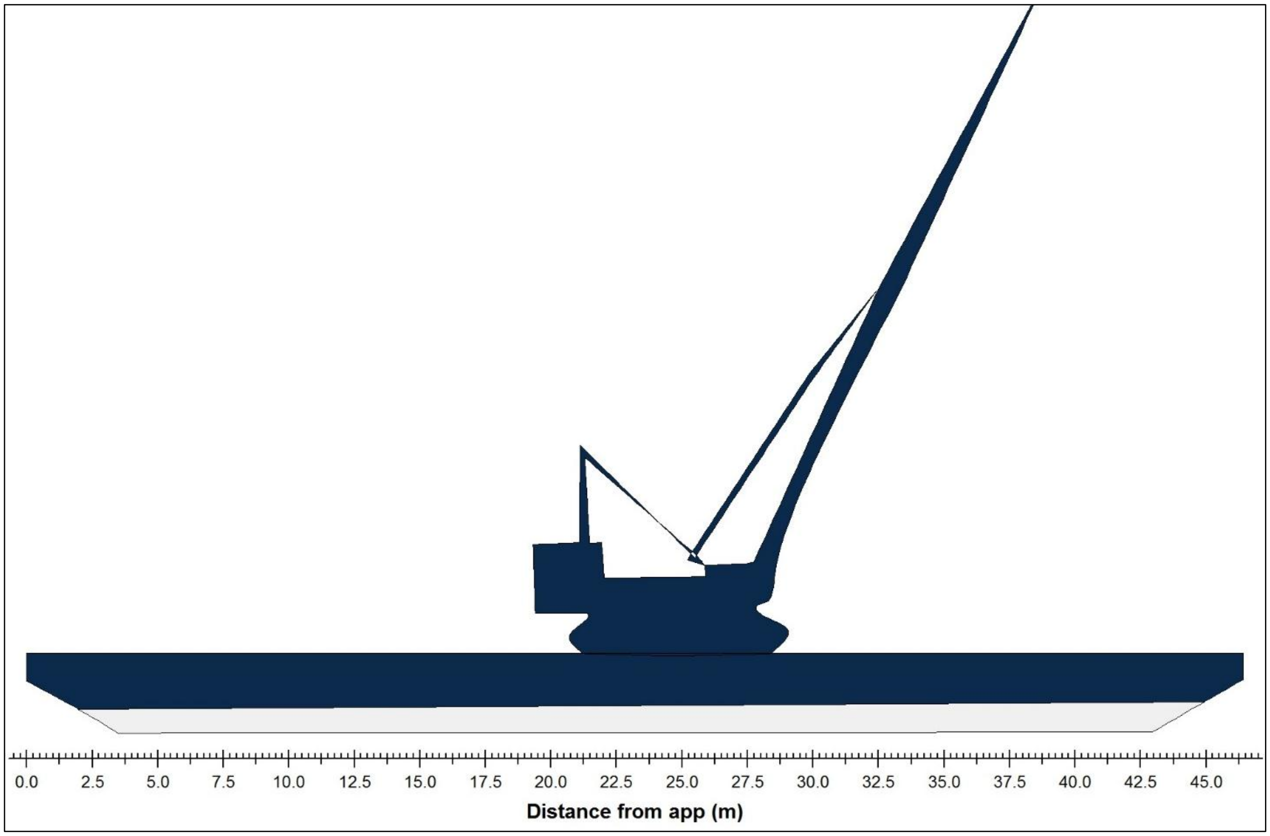


Stability curve
Heeling moment
Steady wind lever
Wind gust lever
Area 1.7-30.0=1.215
Area B=90.040
Angle of heel under action of steady wind=1.8°
Gust equilibrium=1.8°
3.4°
Rollback angle=-51.1°
Area A=163.198
Righting lever (m)
Heeling angle (°)
3.00
2.50
2.00
1.50
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-3.50
-4.00
-4.50
-60
-50
-40
-30
-20
-10
0
10
20
30
40
50
60

Evaluation of criteria						
Minimum design criteria applicable to all ships						
Description	**Attained value**		**Criterion**	**Required value**		**Complies**
Area 0º - 30º	1.2146	*(mrad)*	>=	0.0800	*(mrad)*	**YES**
Severe wind and rolling criterion						**YES**
Wind pressure	55.1	*(kg/m^2)*				
Wind area	135.82	*(m^2)*				
Steady wind lever	0.034	*(m)*				
Deck immersion angle	13.19	(Degr.)				
Maximum allowed static heeling angle	1.8	*(Degr.)*	<=	16.0	*(Degr.)*	**YES**
Max allowed ratio static angle/deck immersion angle	0.135		<=	0.500		**YES**
Range of stability	48.2	*(Degr.)*	>=	20.0	*(Degr.)*	**YES**
Crane criteria						
Heeling moment	3.4	*(Degr.)*	<	15.0	*(Degr.)*	**YES**
Heeling moment	508.86	*(Tonnes*m)*				
Attained value smaller than deck immersion angle	3.4	*(Degr.)*	<	13.2	*(Degr.)*	**YES**
ratio of area						**YES**
Heeling moment:	Heeling moment					
Lower angle	3.4	*(Degr.)*				
Upper angle	46.0	*(Degr.)*				
Area A	1.1474	*(mrad)*				
Area B	1.6137	*(mrad)*				
Ratio of areaA/areaB	0.711		>=	0.400		**YES**
GZc/GZmax	0.212		<=	0.600		**YES**
The vessel complies with the stability criteria						

11.5 LC-04- LOADING CONDITION WITH CRANE LOCATED AT CENTER AND HOOK LOAD TOWARDS FORWARD

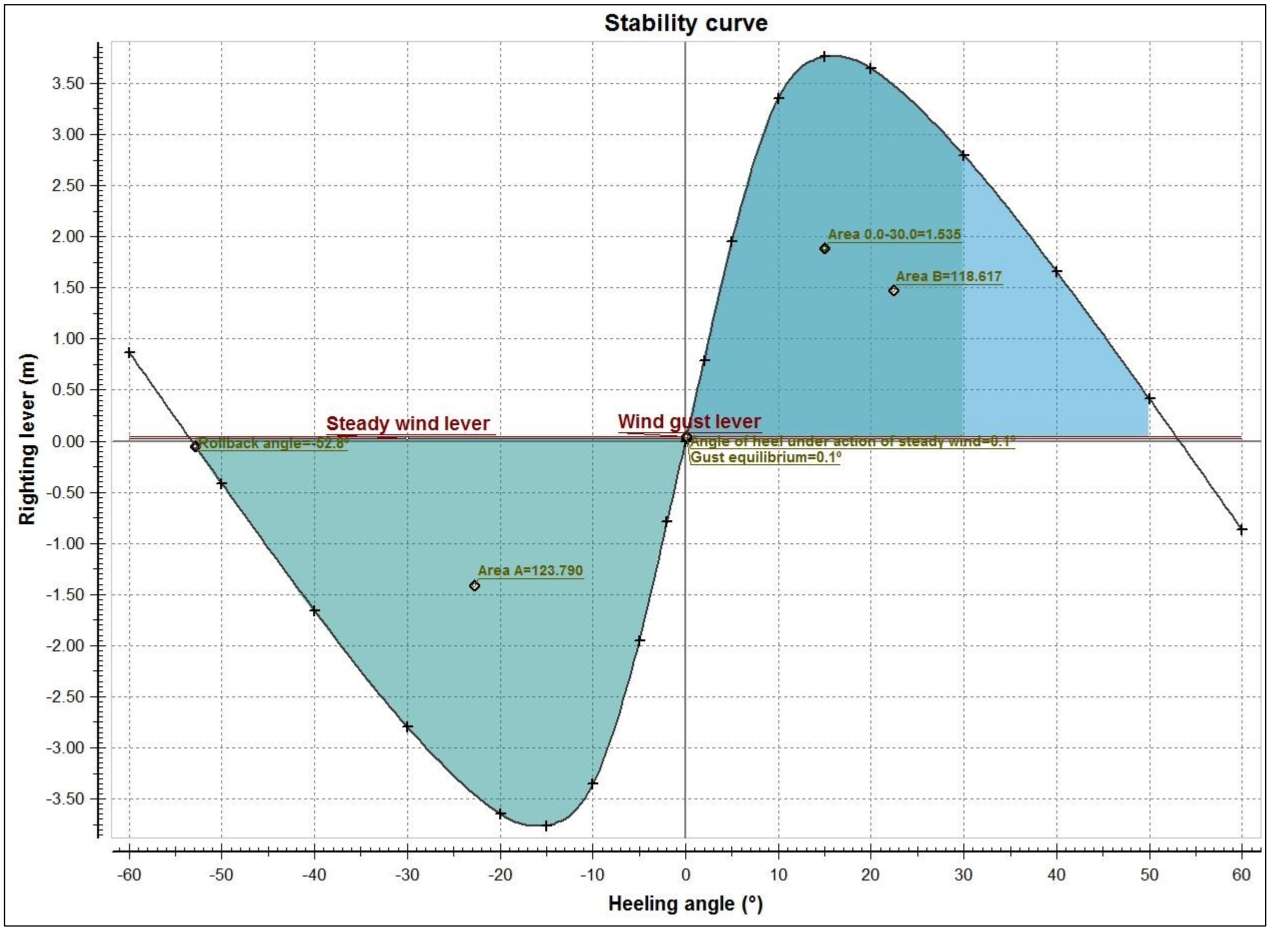
The present load case for the intact stability analysis has been done by considering the Lightship weight of the pontoon, Crane is located at center side and it is operating with a radius of 16.0 m towards forward end from the crane location. It is not recommended to operate the crane with more than the mentioned operating radius for the particular load.



Description	Density (t/m^3)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Miscellaneous							
CRANE			218.00	6.000	23.200	0.000 (CL)	
HOOK LOAD			45.00	39.327	39.200	0.000 (CL)	
Totals for Miscellaneous			**263.00**	**11.702**	**25.938**	**0.000 (CL)**	
Lightship			500.00	3.000	23.200	0.000 (CL)	
Deadweight			263.00	11.702	25.938	0.000 (CL)	0.00
Displacement			763.00	6.000	24.144	0.000 (CL)	0.00

Hydrostatic particulars			
List	0.00 (CL) *(Degr.)*	**KM**	28.548 *(m)*
Draft aft pp	0.868 *(m)*	**GM solid**	22.548 *(m)*
Mean draft	1.015 *(m)*	**GG'**	0.000 *(m)*
Draft forward pp	1.162 *(m)*	**GM liquid**	22.548 *(m)*
Trim	0.294 *(m)*	**MCT**	26.349 *(t*m/cm)*
Volume	744.38 *(m^3)*	**Relative water density**	1.0250
LCF	23.409 *(m)*	**Immersion rate**	7.933 *(t/cm)*

GZ values	
Heeling angle *(Degr.)*	**GZ** *(m)*
0.00 (CL)	0.000
2.00 (PS)	0.788
5.00 (PS)	1.954
10.00 (PS)	3.357
15.00 (PS)	3.760
20.00 (PS)	3.648
30.00 (PS)	2.797
40.00 (PS)	1.660
50.00 (PS)	0.414
60.00 (PS)	-0.868



Evaluation of criteria				
Minimum design criteria applicable to all ships				
Description	**Attained value**	**Criterion**	**Required value**	**Complies**
Area 0º - 30º	1.5352 *(mrad)*	>=	0.0800 *(mrad)*	YES
Severe wind and rolling criterion				YES
Wind pressure	55.1 *(kg/m^2)*			
Wind area	130.94 *(m^2)*			
Steady wind lever	0.029 *(m)*			
Deck immersion angle	12.71 (Degr.)			
Maximum allowed static heeling angle	0.1 *(Degr.)*	<=	16.0 *(Degr.)*	YES
Max allowed ratio static angle/deck immersion angle	0.006	<=	0.500	YES
Range of stability	53.2 *(Degr.)*	>=	20.0 *(Degr.)*	YES
The vessel complies with the stability criteria				

12 RECOMMENDATIONS

1. In LC-02 deck loads without crane Hook load, the vessel attains a heel of ~0.00 degree. It is recommended that vessel should sail without heel. So in sailing condition the deck items should be arranged such that there is no heel for the vessel.
2. In LC-03 the crane is located at the Centre and operates hook load of 45Tons is considered towards port with a crane operating radius of 16m. The vessel attains a heel of 2.39 degree. It is not recommended to operate the crane with more than the mentioned operating radius for the particular load.
3. In LC-04 the crane is located at the center and operates hook load of 45Tons is considered towards port with a crane operating radius of 16 m the vessel attains a heel of 1.69 degree. It is not recommended to operate the crane with more than the mentioned operating radius for the particular load.
4. In LC-05 the crane is located at the center and operates hook load of 45Tons is considered towards forward with a crane operating radius of 16 m the vessel attains a trim of 0.294 m by bow. It is not recommended to operate the crane with more than the mentioned operating radius for the particular load
5. All the deck equipment should be placed carefully, if any change in location of the equipment from its original place care should be taken to avoid heel, whereas little trim by aft is acceptable.
6. Prior to the lashing arrangements for the crane, the position of the crane should be checked such that there is no heel for the vessel.
7. We have assumed the vessel structural condition is suitable to take the loads as given in stowage plan along with the floating environment load.
8. Deck equipment should be properly secured in position during sailing.
9. The openings like hatch and man holes are to be properly closed during sailing.
10. The crane should not be operated in sailing condition and be in lashing position.

13 CONCLUSION

The “46.4 M FLAT TOOP PONTOON” stability assessment is carried out for the loading conditions mentioned in section 11. From the results with this loading conditions and loading pattern Pontoon has adequate stability margin and all the stability criteria are meeting the guideline requirements. From the trim and stability assessment it can be concluded that “46.4 M FLAT TOOP PONTOON” Pontoon with 250 MT self-weight crane with 16 m operating radius with a hook load of 45 MT deck equipment is safe in stability point of view.